

MIT INTEGRATION BEE

SOLUTIONS OF QUALIFYING TESTS
FROM 2010 TO 2023

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FIRST EDITION

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To my parents, my wife, and my lovely son Ali.

Preface

The Integration Bee is an annual integral calculus competition pioneered in 1981 at the Massachusetts Institute of Technology (MIT). Nowadays, Integration Bee contests are regularly conducted in major American universities, including the University of Florida, Florida Polytechnic University, the University of Scranton, Connecticut College, Central Connecticut State University, Columbia University, the State University of New York, the University of Wisconsin–Madison, Prairie State College (Illinois), the University of Illinois at Urbana–Champaign, University of Dayton (Ohio), Louisiana Tech University, the University of North Texas, Brigham Young University, Utah Valley University, Fresno State University, Cosumnes River College, the University of California, Berkeley, various other institutions in California, and Oregon State University. The Integration Bee is also held at the University of Cambridge, University of New South Wales, Australia, and the Indian Institute of Science Education and Research, Pune. The Bee held at Cambridge has participants from other British universities, including the University of Oxford, Imperial College London, Durham University, and the University of Warwick.

While integral calculus is no longer an actively researched topic in mathematics, there is some correlation between success in the integration bee and success in other areas of mathematics, where the Integration Bees share a common goal of challenging students to solve complex integration problems quickly and accurately. They provide opportunities for students to test their mathematical skills and compete with their peers. While the format and rules may vary, the Integration Bee competitions in the United States all share a commitment to promoting mathematical excellence and fostering a love of mathematics in students.

Despite the great popularity of MIT Integration Bee, there is no book that contains sufficient guidance and comprehensive and detailed solutions to the proposed integrals (or questions) each year. This created an incentive to solve these integrals (from 2010 to 2023). The solutions are in different ways, with some useful comments and recommendations. It is worth noting that there are individual efforts by some of those who offer solutions to these integrals on YouTube¹. Also, there are sparse solutions for some different integrals in <https://www.quora.com/>.

¹See for example the following channels: blackpenredpen, Owls Math, Math Solving Channel, Polar Pi, K.O. MATH., MathTuts

The main reason for presenting solutions to problems from 2010 to 2023 only, although the competition started in 1981, is the lack of access to issues that were addressed before 2010. On the official website of the MIT Integration Bee², there are only integrals that were given from 2010 to 2023, with the final answers for each question without any detail to the solution, although there are many challenging problems that require time and advanced calculus skills to solve. Also, while teaching students of all scientific disciplines the course of calculus, the search for various problems and integrals with multiple levels of challenges formed a great incentive to collect and solve these problems. The book will also be a good reference for teachers who teach calculus, as it provides them with many problems of different levels of difficulty, which are fun for students while teaching calculus. Therefore, this book is for students preparing for the Integration Bee, teachers looking for integration problems with different levels of challenge to use in their classroom, or math enthusiasts looking to expand their knowledge and skills in integration calculus, where this book has something for you.

This book consists of 15 chapters. In the first chapter, the most important ideas and techniques of integration were presented. We mentioned the basic integration formulas, integration by substitution, integration by parts, trigonometric and hyperbolic integrals, integrals of irrational functions, integration of binomial differentials, the definition of beta and gamma functions, and some additions such as the King property of integration, the Gaussian integral, Leibniz integral rule and Lebesgue's dominated convergence theorem. In the second chapter, the integrals that were given in the competition MIT Integration Bee from 2010 to 2023 were presented. In the remaining chapters, detailed solutions to the integrals of each year were presented in their own chapter.

I want to mention that this is the first edition of the book, which will be updated periodically every year, as solutions to problems will be added for the coming years. Finally, I would like to thank all of those who supported me in any respect during my work on this book. Many thanks to Ali Shadhar who provide me with suitable Latex files for publishing the book on Amazon, without his help it was very hard to publish this book with good quality. I would be grateful to everyone who will find any typos, addition, comment, or other remarkable methods to the solution and write me to mohammad.alkousa@phystech.edu. I would also be very grateful to everyone who can send questions that were presented in the competition before 2010 so that they can be solved in detail and included in another separate book.

Mohammad S. Alkousa
May 2023

²<https://math.mit.edu/~yyaol/integrationbee.html>

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Chapter 1

Review of the Fundamentals and Some Techniques of Integration

In this chapter, we will mention the fundamental rules and some techniques of integration, which are widely used in integration arithmetic competitions. For more details, there are many books and references on calculus, you can see for example [1, 2, 3].

1.1 Indefinite integrals, basic integration formulas

Definition 1 Let F be a differentiable function on a given open interval I , and f be a function defined on I . The function F is called an antiderivative of a function f on a given open interval if $F'(x) = f(x)$ for all $x \in I$.

Theorem 1 If $F(x)$ is any antiderivative of $f(x)$ on an open interval I , then for any constant C the function $F(x) + C$ is also an antiderivative on I . Moreover, each antiderivative of $f(x)$ on I can be expressed in the form $F(x) + C$ by choosing the constant C appropriately.

The process of finding antiderivatives is called *antidifferentiation* or *integration*. Thus, if

$$\frac{d}{dx}F(x) = f(x),$$

then integrating (or antidifferentiating) the function $f(x)$ produces an antiderivative of the form $F(x) + C$, and we write

$$\int f(x)dx = F(x) + C,$$

where $C \in \mathbb{R}$ is an arbitrary constant.

1.1.1 Basic integration formulas

Here and in what follows, $C \in \mathbb{R}$ denotes any arbitrary constant.

(1)

$$\int a dx = ax + C, \quad a \in \mathbb{R}.$$

(2)

$$\int x^a dx = \frac{x^{a+1}}{a+1} + C, \quad a \in \mathbb{R} \setminus \{-1\}.$$

More generally, we have

$$\int (f(x))^a f'(x) dx = \frac{(f(x))^{a+1}}{a+1} + C, \quad a \in \mathbb{R} \setminus \{-1\}.$$

(3)

$$\int \frac{dx}{x} = \ln |x| + C, \quad x \neq 0.$$

More generally, we have

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C, \quad f(x) \neq 0.$$

(4)

$$\int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + C, \quad a \in \mathbb{R}^*, b \in \mathbb{R}.$$

(5)

$$\int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + C, \quad a \in \mathbb{R}^*, b \in \mathbb{R}.$$

(6)

$$\begin{aligned} \int \frac{dx}{\cos^2(ax + b)} &= \int \sec^2(ax + b) dx \\ &= \frac{1}{a} \tan(ax + b) + C, \quad a \in \mathbb{R}^*, b \in \mathbb{R}. \end{aligned}$$

(7)

$$\begin{aligned} \int \frac{dx}{\sin^2(ax + b)} &= \int \csc^2(ax + b) dx \\ &= -\frac{1}{a} \cot(ax + b) + C, \quad a \in \mathbb{R}^*, b \in \mathbb{R}. \end{aligned}$$

$$(8) \quad \int \sinh(ax + b) dx = \frac{1}{a} \cosh(ax + b) + C, \quad a \in \mathbb{R}^*, b \in \mathbb{R}.$$

$$(9) \quad \int \cosh(ax + b) dx = \frac{1}{a} \sinh(ax + b) + C, \quad a \in \mathbb{R}^*, b \in \mathbb{R}.$$

$$(10) \quad \int \frac{dx}{\cosh^2(ax + b)} = \frac{1}{a} \tanh(ax + b) + C, \quad a \in \mathbb{R}^*, b \in \mathbb{R}.$$

$$(11) \quad \int \frac{dx}{\sinh^2(ax + b)} = -\frac{1}{a} \coth(ax + b) + C, \quad a \in \mathbb{R}^*, b \in \mathbb{R}.$$

$$(12) \quad \int a^{\alpha x + \beta} dx = \frac{a^{\alpha x + \beta}}{\alpha \ln(a)} + C, \quad a \in]0, \infty[\setminus \{1\}, \alpha \in \mathbb{R}^*, \beta \in \mathbb{R}.$$

Special case: when $a = e$, then we get

$$\int e^{\alpha x + \beta} dx = \frac{1}{\alpha} e^{\alpha x + \beta} + C.$$

$$(13) \quad \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad a \in \mathbb{R}^*.$$

$$(14) \quad \int \frac{dx}{a^2 - x^2} = \frac{1}{a} \operatorname{artanh}\left(\frac{x}{a}\right) + C = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C, \quad a \in \mathbb{R}^*.$$

$$(15) \quad \int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C = -\arccos\left(\frac{x}{a}\right) + C, \quad x < a.$$

$$(16) \quad \int \frac{dx}{\sqrt{x^2 + a^2}} = \ln \left| x + \sqrt{x^2 + a^2} \right| + C = \operatorname{arsinh}\left(\frac{x}{a}\right) + C, \quad a > 0.$$

$$(17) \quad \int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left| x + \sqrt{x^2 - a^2} \right| + C = \operatorname{arcosh}\left(\frac{x}{a}\right) + C, \quad x > a > 0.$$

(18)

$$\int \frac{dx}{\cos x} = \int \sec x \, dx = \ln |\sec x + \tan x| + C = \ln \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + C.$$

(19)

$$\int \frac{dx}{\sin x} = \int \csc x \, dx = -\ln |\csc x + \cot x| + C = \ln \left| \tan \left(\frac{x}{2} \right) \right| + C.$$

(20) Let $a, b \in \mathbb{R}$, such that $a^2 + b^2 \neq 0$. Then

$$I := \int e^{ax} \cos(bx) \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos(bx) + b \sin(bx)) + C, \quad (1.1)$$

where $C \in \mathbb{R}$.

Indeed, by using the integration by parts (see Section 1.4), let us assume

$$u = e^{ax} \implies du = ae^{ax} \, dx, \quad dv = \cos(bx) \, dx \implies v = \frac{1}{b} \sin(bx).$$

Thus, we have

$$I = \frac{1}{b} e^{ax} \sin(bx) - \underbrace{\frac{a}{b} \int e^{ax} \sin(bx) \, dx}_{:=J}.$$

For the integral J , by using the integration by parts, let us assume

$$u = e^{ax} \implies du = ae^{ax} \, dx, \quad dv = \sin(bx) \, dx \implies v = -\frac{1}{b} \cos(bx).$$

Thus, we have

$$\begin{aligned} I &= \frac{1}{b} e^{ax} \sin(bx) - \frac{a}{b} \left(-\frac{1}{b} e^{ax} \cos(bx) + \frac{a}{b} \underbrace{\int e^{ax} \cos(bx) \, dx}_I \right) \\ &= \frac{1}{b} e^{ax} \sin(bx) + \frac{a}{b^2} e^{ax} \cos(bx) - \frac{a^2}{b^2} I. \end{aligned}$$

Thus, we get

$$\left(1 + \frac{a^2}{b^2} \right) I = e^{ax} \left(\frac{b \sin(bx) + a \cos(bx)}{b^2} \right).$$

Therefore, we have

$$I = \frac{e^{ax}}{a^2 + b^2} (a \cos(bx) + b \sin(bx)) + C.$$

(21) In similar of the previous, we get

$$\int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \sin(bx) - b \cos(bx)) + C, \quad (1.2)$$

where $C \in \mathbb{R}$.

1.1.2 Properties of indefinite integrals

(1)

$$\left(\int f(x) dx \right)' = f(x).$$

(2)

$$d \left(\int f(x) dx \right) = f(x) dx.$$

(3)

$$\int dF(x) = F(x) + C, \quad C \in \mathbb{R}.$$

(4) Let $f_1, \dots, f_n$ be integrable functions, then

$$\int (f_1(x) + \dots + f_n(x)) dx = \int f_1(x) dx + \dots + \int f_n(x) dx.$$

(5)

$$\int a f(x) dx = a \int f(x) dx, \quad \forall a \in \mathbb{R}^*.$$

1.2 Definite integrals

Theorem 2 (The Fundamental Theorem of Calculus, Part 1) *If f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$ (i.e. $F'(x) = f(x)$, $\forall x \in [a, b]$), then*

$$\int_a^b f(x) dx = \int_a^b F'(x) dx = F(b) - F(a) := [F(x)]_a^b.$$

Theorem 3 *If a function f is continuous on an interval $[a, b]$, then f is integrable on $[a, b]$, and the area A between the graph of f and the interval $[a, b]$ is*

$$A = \int_a^b f(x)dx.$$

1.2.1 Properties of definite integrals

- (1) If a is in the domain of f , then

$$\int_a^a f(x)dx = 0.$$

- (2) If f is integrable on $[a, b]$, then

$$\int_a^b f(x)dx = - \int_b^a f(x)dx.$$

- (3) If f is integrable on $[a, b]$ and if $\alpha \in \mathbb{R}$ is a constant, then

$$\int_a^b \alpha f(x)dx = \alpha \int_a^b f(x)dx.$$

- (4) If $f_1, \dots, f_n$ are integrable functions on $[a, b]$, then

$$\int_a^b (f_1(x) + \dots + f_n(x)) dx = \int_a^b f_1(x)dx + \dots + \int_a^b f_n(x)dx.$$

- (5) If f is integrable on a closed interval containing the three points a, b, c such that $a \leq c \leq b$, then

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

- (6) The variable of integration in a definite integral plays no role in the end result, it is often referred to as a *dummy variable*. This means

$$\int_a^b f(x)dx = \int_a^b f(t)dt.$$

1.3 Integration by substitution

The substitution formula is often written in the form

$$\int f(u)du = \int f(g(x))g'(x)dx, \quad \text{where } u = g(x). \quad (1.3)$$

For definite integrals, the formula (1.3), will has the following form

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du. \quad (1.4)$$

1.4 Integration by parts

Be attentive!

$$\int f(x)g(x)dx \neq \int f(x)dx \int g(x)dx.$$

But, if f' and g' are continuous functions¹ of x , then the integration by parts formula has the following form

$$\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx. \quad (1.5)$$

Sometimes it is easier to remember the formula (1.5) if we write it in differential form. Let $u = f(x)$ and $v = g(x)$. Then $du = f'(x)dx$ and $dv = g'(x)dx$. Using the substitution rule, the integration by parts formula (1.5) becomes

$$\int u dv = uv - \int v du. \quad (1.6)$$

This formula expresses one integral $\int u dv$, in terms of a second integral $\int v du$. With a proper choice of u and v , the second integral may be easier to evaluate than the first. In using the formula (1.6), various choices may be available for u and dv .

For the definite integrals, we have

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du.$$

From (1.6), we can see that the Integration by parts is a technique for simplifying integrals of the multiplication of two functions. It is useful when one of these functions can be differentiated repeatedly and the second can be integrated repeatedly without difficulty.

¹See: Alexander Kheifets and James Propp: A Counterexample to Integration by Parts. Mathematics Magazine, Vol. 83, No. 3 (June 2010), pp. 222-225. Where in this paper, the authors exhibit two differentiable functions f and g for which the function $f'g$ and fg' are not integrable, so that the integration by parts formula does not apply.

Let us mention some of the integrals that can be calculated by using the integration by parts.

(1) The integrals

$$\int P_n(x) \sin(ax + b) dx, \quad \int P_n(x) \cos(ax + b) dx, \quad \int P_n(x) e^{ax+b} dx,$$

$$\int P_n(x) \sinh(ax + b) dx, \quad \int P_n(x) \cosh(ax + b) dx,$$

where $P_n(x)$ is a polynomial of degree n and $a, b \in \mathbb{R}$.

For these integrals, we assume $u = P_n(x)$ and the remainder is dv . By using the integration by parts n times we calculate the given integral.

(2) The integrals

$$\int P_n(x) \ln^m x dx, \quad \int \arctan x dx, \quad \int \operatorname{arccot} x dx, \quad \int \arcsin x dx,$$

$$\int \arccos x dx, \quad \int \operatorname{arsinh} x dx, \quad \int \operatorname{arcosh} x dx.$$

For the first integral, we assume $u = \ln^m x$ and by using the integration by parts m times we calculate this integral. For other integrals, we assume $dv = dx$ and the remainder (i.e. the inverse trigonometric or inverse hyperbolic functions) is u .

1.5 Trigonometric and hyperbolic integrals

1.5.1 Basic identities for trigonometric and inverse trigonometric functions

Remember that, for any $x \in \mathbb{R}$ and $k \in \mathbb{Z}$, we have

$$\begin{aligned} \sin(-x) &= -\sin x, & \cos(-x) &= \cos x, \\ \tan(-x) &= -\tan x, & \cot(-x) &= -\cot x. \\ \sin(x + 2\pi k) &= \sin x, & \cos(x + 2\pi k) &= \cos x, \\ \tan(x + \pi k) &= \tan x, & \cot(x + \pi k) &= \cot x. \end{aligned}$$

Double-angle formulae

$$\sin(2x) = 2 \sin x \cos x = \frac{2 \tan x}{1 + \tan^2 x},$$

$$\cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x = \frac{1 - \tan^2 x}{1 + \tan^2 x},$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}.$$

Sum and difference formulas

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y,$$

$$\cos(x \mp y) = \cos x \cos y \pm \sin x \sin y,$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}.$$

Formulas for transforming the product into sum or difference

$$\cos(mx) \cos(nx) = \frac{1}{2} [\cos(m+n)x + \cos(m-n)x],$$

$$\sin(mx) \cos(nx) = \frac{1}{2} [\sin(m+n)x + \sin(m-n)x],$$

$$\sin(mx) \sin(nx) = -\frac{1}{2} [\cos(m+n)x - \cos(m-n)x].$$

Inverse trigonometric functions

For the domains and ranges of the inverse trigonometric functions, we have

$$\arcsin : [-1, 1] \longrightarrow [-\pi/2, \pi/2], \quad \arccos : [-1, 1] \longrightarrow [0, \pi],$$

$$\arctan : \mathbb{R} \longrightarrow (-\pi/2, \pi/2), \quad \operatorname{arccot} : \mathbb{R} \longrightarrow (0, \pi).$$

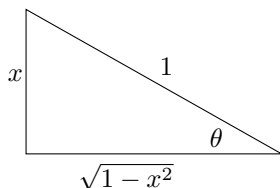
For these functions, we have

$$\arcsin(-x) = -\arcsin x, \quad \arccos(-x) = \pi - \arccos x,$$

$$\arctan(-x) = -\arctan x, \quad \operatorname{arccot}(-x) = \pi - \operatorname{arccot} x.$$

Relationships between trigonometric functions and inverse trigonometric functions

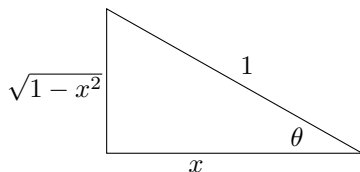
(1) If $\theta = \arcsin x$, where $x \in [-1, 1]$, then we can get the following figure



and

$$\sin(\arcsin x) = x, \quad \cos(\arcsin x) = \sqrt{1-x^2}, \quad \tan(\arcsin x) = \frac{x}{\sqrt{1-x^2}}.$$

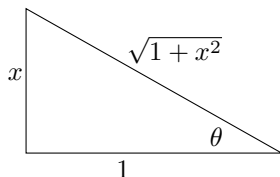
(2) If $\theta = \arccos x$, where $x \in [-1, 1]$, then we can get the following figure



and

$$\sin(\arccos x) = \sqrt{1-x^2}, \quad \cos(\arccos x) = x, \quad \tan(\arccos x) = \frac{\sqrt{1-x^2}}{x}.$$

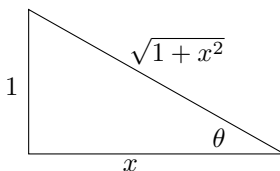
(3) If $\theta = \arctan x$, where $x \in \mathbb{R}$, then we can get the following figure



and

$$\sin(\arctan x) = \frac{x}{\sqrt{1+x^2}}, \quad \cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}, \quad \tan(\arctan x) = x.$$

(4) If $\theta = \operatorname{arccot} x$, then we can get the following figure



and

$$\sin(\operatorname{arccot} x) = \frac{1}{\sqrt{1+x^2}} \quad \forall x \in \mathbb{R}, \quad \cos(\operatorname{arccot} x) = \frac{x}{\sqrt{1+x^2}} \quad \forall x \in \mathbb{R},$$

$$\tan(\operatorname{arccot} x) = \frac{1}{x} \quad \forall x \in \mathbb{R} \setminus \{0\}.$$

1.5.2 Trigonometric integrals

Let R be a rational function in its arguments (maybe more than one variable). For integrals

$$\int R(\sin x, \cos x) dx, \quad (1.7)$$

we can use the substitution $\tan\left(\frac{x}{2}\right) = t$, thus $x = 2 \arctan t$, and $dx = \frac{2dt}{1+t^2}$. Also, we have

$$\begin{aligned} \sin x &= \frac{2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)}{1} = \frac{2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)}{\sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right)} = \frac{2 \tan\left(\frac{x}{2}\right)}{1 + \tan^2\left(\frac{x}{2}\right)} = \frac{2t}{1+t^2}, \\ \cos x &= \frac{\cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)}{1} = \frac{\cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)}{\cos^2\left(\frac{x}{2}\right) + \sin^2\left(\frac{x}{2}\right)} = \frac{1 - \tan^2\left(\frac{x}{2}\right)}{1 + \tan^2\left(\frac{x}{2}\right)} = \frac{1-t^2}{1+t^2}. \end{aligned}$$

This gives the following integral

$$\int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2dt}{1+t^2},$$

which is an integral of a rational function in t .

For the trigonometric integrals (1.7), there are some special cases that can be listed as follows.

(1) For the integrals

$$\int R(\cos x) \sin x dx,$$

we can use the substitution $\cos x = t$, so that $-\sin x dx = dt$. This gives the integral $-\int R(t)dt$, which is an integral of the rational function in t .

Example 1 For the integral

$$I = \int \frac{\sin^3 x}{2 + \cos x} dx,$$

we have

$$I = \int \frac{\sin^2 x \sin x}{2 + \cos x} dx = \int \frac{1 - \cos^2 x}{2 + \cos x} \sin x dx.$$

Let us use the substitution $\cos x = t$, which gives the following

$$I = - \int \frac{1-t^2}{1+t^2} dt = \int \left(t - 2 + \frac{3}{2+t} \right) dt = \frac{t^2}{2} - 2t + 3 \ln |2+t| + C$$

$$= \frac{1}{2} \cos^2 x - \cos x + 3 \ln |2 + \cos x| + C,$$

where $C \in \mathbb{R}$.

(2) For the integrals

$$\int R(\sin x) \cos x dx,$$

we can use the substitution $\sin x = t$, so that $\cos x dx = dt$. This gives the integral $\int R(t) dt$, which is an integral of the rational function in t .

(3) For the integrals

$$\int R(\tan x) dx,$$

we can use the substitution $\tan x = t$, so that $x = \arctan t$ and $dx = \frac{dt}{1+t^2}$. This gives the integral $\int R(t) \frac{dt}{1+t^2}$, which is an integral of the rational function in t .

(4) For the integrals

$$\int R(\sin^m x, \cos^n x) dx,$$

where the powers m, n of $\sin x$ and $\cos x$ are even, we can use the substitution

$$\tan x = t \implies x = \arctan t \implies dx = \frac{dt}{1+t^2},$$

and

$$\cos^2 x = \frac{1}{1 + \tan^2 x} = \frac{1}{1 + t^2}, \quad \sin^2 x = \frac{\tan^2 x}{1 + \tan^2 x} = \frac{t^2}{1 + t^2}.$$

This gives an integral of a rational function in t .

Example 2 For the integral

$$I = \int \frac{dx}{2 - \sin^2 x},$$

with $\tan x = t$, we have

$$\begin{aligned} I &= \int \frac{dt}{(1+t^2) \left(2 - \frac{t^2}{1+t^2}\right)} = \int \frac{dt}{2+t^2} = \frac{1}{\sqrt{2}} \arctan \left(\frac{t}{\sqrt{2}} \right) + C \\ &= \frac{1}{\sqrt{2}} \arctan \left(\frac{\tan x}{\sqrt{2}} \right) + C, \end{aligned}$$

where $C \in \mathbb{R}$.

(5) For the integrals

$$\int \sin^m x \cos^n x dx \quad (m, n \in \mathbb{Z}), \quad (1.8)$$

we can list the following cases.

- One of the numbers m, n is an odd positive (for example n is this number). Then by setting $n = 2p + 1$, where $p \in \mathbb{Z}^+$, we get

$$\begin{aligned} I &= \int \sin^m x \cos^{2p+1} x dx = \int \sin^m x \cos^{2p} x \cos x dx \\ &= \int \sin^m x (1 - \sin^2 x)^p \cos x dx. \end{aligned}$$

Thus, when we use the substitution $\sin x = t$, we get the following integral

$$I = \int t^m (1 - t^2)^p dt,$$

which is an integral of a rational function when $m < 0$. If $m > 0$, then the integrand will be a polynomial with degree $m + 2p$.

Example 3 For the integral

$$I = \int \frac{\cos^3 x}{\sin^4 x} dx,$$

we have

$$I = \int \frac{\cos^2 x \cos x}{\sin^4 x} dx = \int \frac{(1 - \sin^2 x)}{\sin^4 x} \cos x dx.$$

Let us use the substitution $\sin x = t$, thus $\cos x dx = dt$. This gives the following

$$\begin{aligned} I &= \int \frac{1 - t^2}{t^4} dt = \int (t^{-4} - t^{-2}) dt = -\frac{1}{3t^3} + \frac{1}{t} + C \\ &= -\frac{1}{3 \sin^3 x} + \frac{1}{\sin x} + C, \end{aligned}$$

where $C \in \mathbb{R}$.

- Both numbers m and n are even. Let $m = 2p$ and $n = 2q$. We have

$$\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos(2x), \quad \cos^2 x = \frac{1}{2} + \frac{1}{2} \cos(2x). \quad (1.9)$$

Thus, we have the following integral

$$\int \sin^{2p} x \cos^{2q} x dx = \int \left(\frac{1}{2} - \frac{1}{2} \cos(2x) \right)^p \left(\frac{1}{2} + \frac{1}{2} \cos(2x) \right)^q dx.$$

After doing some simple calculations, we get even or odd powers for $\cos(2x)$. For the terms with odd powers, we return to the first case. For the terms with even powers, we can use (1.9) which gives an integral of the form $\int \cos(\alpha x) dx$, where $\alpha \in \mathbb{R}^*$.

Example 4 For the integral

$$I = \int \sin^4 x dx,$$

we have

$$\begin{aligned} I &= \int \sin^4 x dx = \frac{1}{4} \int (1 - \cos(2x))^2 dx \\ &= \frac{1}{4} \int (1 - 2\cos(2x) + \cos^2(2x)) dx \\ &= \frac{1}{4} \int \left(1 - 2\cos(2x) + \frac{1}{2} + \frac{1}{2} \cos(4x) \right) dx \\ &= \frac{1}{4} \int \left(\frac{3}{2} - 2\cos(2x) + \frac{1}{2} \cos(4x) \right) dx \\ &= \frac{3}{8}x - \frac{1}{4} \sin(2x) + \frac{1}{32} \cos(4x) + C, \end{aligned}$$

where $C \in \mathbb{R}$.

- Both numbers m, n are even and one of them is negative. In this case, we can use the substitution $\tan x = t$.

Example 5 For the integral

$$I = \int \frac{\sin^2 x}{\cos^6 x} dx,$$

we have

$$I = \int \frac{\sin^2 x}{\cos^2 x} \frac{1}{\cos^2 x} \frac{dx}{\cos^2 x}.$$

Let us use the substitution $\tan x = t$, thus $\frac{dx}{\cos^2 x} = dt$. This gives the following

$$I = \int t^2(1 + t^2)dt = \int (t^4 + t^2)dt = \frac{t^5}{5} + \frac{t^3}{3} + C$$

$$= \frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} + C,$$

where $C \in \mathbb{R}$.

1.5.3 Hyperbolic integrals

Hyperbolic integrals can be calculated in a similar way as trigonometric integrals. Here we mention some useful formulas for hyperbolic functions.

Definitions

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2},$$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad \coth x = \frac{1}{\tanh x} = \frac{\sinh x}{\cosh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}.$$

Fundamental formulas

$$\cosh^2 x - \sinh^2 x = 1, \quad \sinh(-x) = -\sinh x,$$

$$\cosh(-x) = \cosh x, \quad \tanh(-x) = -\tanh x,$$

$$\sinh(a \pm b) = \sinh a \cosh b \pm \cosh a \sinh b,$$

$$\cosh(a \pm b) = \cosh a \cosh b \pm \sinh a \sinh b,$$

$$\tanh(a \pm b) = \frac{\tanh a \pm \tanh b}{1 \pm \tanh a \tanh b},$$

$$\cosh(2x) = \cosh^2 x + \sinh^2 x = 2 \sinh^2 x + 1 = 2 \cosh^2 x - 1,$$

$$\cosh^2 x = \frac{\cosh(2x) + 1}{2}, \quad \sinh^2 x = \frac{\cosh(2x) - 1}{2},$$

$$\sinh(2x) = 2 \sinh x \cosh x, \quad \tanh(2x) = \frac{2 \tanh x}{1 + \tanh^2 x}.$$

Formulas for transforming the product into sum or difference

$$\sinh(mx) \cosh(nx) = \frac{1}{2} [\sinh(m+n)x + \sinh(m-n)x],$$

$$\cosh(mx) \cosh(nx) = \frac{1}{2} [\cosh(m+n)x + \cosh(m-n)x],$$

$$\sinh(mx) \sinh(nx) = -\frac{1}{2} [\cosh(m+n)x - \cosh(m-n)x].$$

Inverse hyperbolic functions

$$\operatorname{arsinh} x = \ln \left(x + \sqrt{x^2 + 1} \right); \quad \forall x \in \mathbb{R},$$

$$\operatorname{arcosh} x = \ln \left(x + \sqrt{x^2 - 1} \right); \quad \forall x \in [1, \infty[,$$

$$\begin{aligned}\operatorname{artanh} x &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right); & \forall x \in]-1, 1[, \\ \operatorname{arcoth} x &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right); & \forall x \in]-\infty, -1[\cup]1, \infty[.\end{aligned}$$

Also, we have the following

$$\begin{aligned}\sinh(\operatorname{arcosh} x) &= \sqrt{x^2 - 1}; & |x| > 1, \\ \sinh(\operatorname{artanh} x) &= \frac{x}{\sqrt{1-x^2}}; & -1 < x < 1, \\ \cosh(\operatorname{arsinh} x) &= \sqrt{1+x^2}; & x \in \mathbb{R}, \\ \cosh(\operatorname{artanh} x) &= \frac{1}{\sqrt{1-x^2}}; & -1 < x < 1, \\ \tanh(\operatorname{arsinh} x) &= \frac{x}{\sqrt{1+x^2}}; & x \in \mathbb{R}, \\ \tanh(\operatorname{arcosh} x) &= \frac{\sqrt{x^2-1}}{x}; & |x| > 1.\end{aligned}$$

1.5.4 Trigonometric and hyperbolic substitutions

Remember that

$$\sin^2 x + \cos^2 x = 1, \quad \cosh^2 x - \sinh^2 x = 1, \quad 1 + \tan^2 x = \frac{1}{\cos^2 x} = \sec^2 x.$$

Therefore

- To calculate

$$\int R\left(x, \sqrt{a^2 - b^2 x^2}\right) dx,$$

where $a, b \in \mathbb{R}^*$ and $R(\cdot, \cdot)$ is a rational function in its arguments, we can use the substitution $x = \frac{a}{b} \sin t$, or $x = \frac{a}{b} \cos t$, or $x = \frac{a}{b} \tanh t$.

- To calculate

$$\int R\left(x, \sqrt{a^2 + b^2 x^2}\right) dx,$$

where $a, b \in \mathbb{R}^*$, and $R(\cdot, \cdot)$ is a rational function in its arguments, we can use the substitution $x = \frac{a}{b} \tan t$, or $x = \frac{a}{b} \sinh t$.

- To calculate

$$\int R\left(x, \sqrt{b^2 x^2 - a^2}\right) dx,$$

where $a, b \in \mathbb{R}^*$, and $R(\cdot, \cdot)$ is a rational function in its arguments, we can use the substitution $x = \frac{a}{b} \sec t$, or $x = \frac{a}{b} \cosh t$.

1.6 Integrals of irrational functions

For the integrals

$$\int R\left(x, \left(\frac{ax+b}{cx+d}\right)^{\frac{m}{n}}, \dots, \left(\frac{ax+b}{cx+d}\right)^{\frac{r}{s}}\right) dx,$$

where $m, n, \dots, r, s \in \mathbb{Z}^*$, R is a rational function in its arguments, and $a, b, c, d \in \mathbb{R}$ such that $ad - bc \neq 0$. We can use the substitution $\frac{ax+b}{cx+d} = t^k$, where k is a common denominator of the denominators $n, \dots, s$. This substitution yields an integral of a rational function.

Special case 1: The integrals that have the following form

$$\int R\left(x, x^{\frac{m}{n}}, \dots, x^{\frac{r}{s}}\right) dx.$$

Special case 1: The integrals that have the following form

$$\int R\left(x, (ax+b)^{\frac{m}{n}}, \dots, (ax+b)^{\frac{r}{s}}\right) dx.$$

For the integrals

$$I = \int \frac{dx}{(x-\alpha)^k \sqrt{ax^2+bx+c}}. \quad (1.10)$$

we use the substitution $\frac{1}{x-\alpha} = t$, so that

$$dx = -\frac{dt}{t^2}, \quad ax^2+bx+c = \frac{(a\alpha^2+b\alpha+c)t^2 + (2a\alpha+b)t + a}{t^2}.$$

This gives the following integral

$$I = - \int \frac{t^{k-1}}{\sqrt{(a\alpha^2+b\alpha+c)t^2 + (2a\alpha+b)t + a}} dt,$$

which can be calculated simpler than the given integral (1.10).

1.7 Integration of binomial differentials

An expression of the form $x^m (a + bx^n)^p$, where $a, b \in \mathbb{R}^*$ and $m, n, p \in \mathbb{Q}$ is called a binomial differential.

Theorem 4 *The integral of a binomial differential*

$$I = \int x^m (a + bx^n)^p dx,$$

is reduced to an integral of a rational function and thus is expressed in terms of elementary functions in the following three cases

1. When $p \in \mathbb{Z}$. In this case, to calculate I , we use the substitution $x = t^k$, where k is a common denominator of the fractions m, n .
2. When $\frac{m+1}{n} \in \mathbb{Z}$. In this case, to calculate I , we use the substitution $a + bx^n = t^r$, where r is the denominator of p .
3. When $\frac{m+1}{n} + p \in \mathbb{Z}$. In this case, we can write

$$I = \int x^m x^{np} (b + ax^{-n})^p dx = \int x^{m+np} (b + ax^{-n})^p dx. \quad (1.11)$$

Let $m_1 := m + np$, $n_1 := -n$, $p_1 := p$. We note that

$$\frac{m_1 + 1}{n_1} = \frac{m + np + 1}{-n} = -\left(\frac{m + 1}{n} + p\right) \in \mathbb{Z}.$$

Thus, the integral (1.11), is an integral of a binomial differential from the second case.

1.8 Beta and Gamma functions

The gamma and beta functions are two important mathematical functions that arise in various areas of mathematics, physics, and engineering. In this section, we mention the definitions of these functions.

1.8.1 Gamma function

This function is also called *Euler's gamma function* and is defined as follows

$$\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, \quad \forall a > 0. \quad (1.12)$$

For this function we have $\Gamma(a+1) = a\Gamma(a)$, and from this relation we get $\Gamma(n+1) = n!$, $\forall n \in \mathbb{N}$. Therefore, we see that the gamma function extends the factorial function from the non-negative integers to other real values.

From (1.12) and Gaussian integral (see (1.14)), we find the following important number $\Gamma(1/2) = \sqrt{\pi}$.

1.8.2 Beta function

For any $a > 0$ and $b > 0$, the beta function (also called the *Euler integral of the first kind*) is defined as follows

$$\beta(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1} dx.$$

The close relationship to the gamma function is

$$\beta(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

The trigonometric form of the beta function is

$$\beta(a, b) = 2 \int_0^{\pi/2} (\sin x)^{2a-1} (\cos x)^{2b-1} dx.$$

1.9 Additional

1.9.1 King property of integration

This property states

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx. \quad (1.13)$$

1.9.2 The Gaussian integral

The Gaussian integral (or *Euler–Poisson integral*) has the following form

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}. \quad (1.14)$$

1.9.3 Leibniz integral rule

Leibniz's integral rule provides us a way to differentiate an integral without first calculating the integral in the first place. This rule states the following:

Let $f(x, t)$ be a function such that both $f(x, t)$ and its partial derivative $f_x(x, t) = \frac{\partial}{\partial x} f(x, t)$ are continuous in t and x in some region of the xt -plane, including $a(x) \leq t \leq b(x)$, $x_0 \leq x \leq x_1$, where $x_0, x_1 \in \mathbb{R}$. Also, suppose that the functions $a(x)$ and $b(x)$ are both continuous and both have continuous derivatives for $x_0 \leq x \leq x_1$. Then, for $x_0 \leq x \leq x_1$,

$$\frac{d}{dx} \left(\int_{a(x)}^{b(x)} f(x, t) dt \right) = f(x, b(x))b'(x) - f(x, a(x))a'(x) + \int_{a(x)}^{b(x)} f_x(x, t) dt.$$

In the special case where the functions $a(x) = a$ and $b(x) = b$ are constants with values that do not depend on x , this simplifies to:

$$\frac{d}{dx} \left(\int_a^b f(x, t) dt \right) = \int_a^b f_x(x, t) dt. \quad (1.15)$$

1.9.4 Lebesgue's dominated convergence theorem

This theorem states that [4, 5]: if $f_n : \mathbb{R} \rightarrow \mathbb{R}$ is a sequence of measurable functions such that the pointwise limit $\lim_{n \rightarrow \infty} f_n(x) := f(x)$ exists, and if there exists an integrable function g such that $|f_n(x)| \leq g(x)$ for all n and for all x , then f is integrable and

$$\int_{\mathbb{R}} f(x) dx = \lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n(x) dx.$$

Chapter 2

The Questions of MIT Integration Bee, Qualifying Tests

2.1 2010 MIT Integration Bee, Qualifying Test

$$\boxed{1} \quad I_1 = \int_0^{\pi/2} \sin x \sin(2x) \sin(3x) dx.$$

$$\boxed{2} \quad I_2 = \int_0^{\pi/2} \sin^3(2x) \cos x dx.$$

$$\boxed{3} \quad I_3 = \int (x+1)^2 (x-1)^{1/3} dx.$$

$$\boxed{4} \quad I_4 = \int x \ln \left(1 + \frac{1}{x} \right) dx.$$

$$\boxed{5} \quad I_5 = \int_0^1 \sin^2(\ln x) dx.$$

$$\boxed{6} \quad I_6 = \int \frac{dx}{1 + 3e^x}.$$

$$\boxed{7} \quad I_7 = \int_{\pi/4}^{\pi/3} \frac{dx}{\sin^3 x \cos^5 x}.$$

$$\boxed{8} \quad I_8 = \int_1^{\infty} \frac{dx}{x\sqrt{x^4-1}}.$$

$$\boxed{9} \quad I_9 = \int \frac{dx}{x(x^5+1)}.$$

$$\boxed{10} \quad I_{10} = \int_0^{\pi/4} \sqrt{\tan x} dx.$$

$$\boxed{11} \quad I_{11} = \int_0^1 \frac{\ln(1+x)}{1+x^2} dx.$$

$$\boxed{12} \quad I_{12} = \int_{64}^{729} \frac{\sqrt{x}}{\sqrt{x}-\sqrt[3]{x}} dx.$$

$$\boxed{13} \quad I_{13} = \int x^x(1+\ln x) dx.$$

$$\boxed{14} \quad I_{14} = \int_0^1 x^{13/2} \sqrt{1+x^{5/2}} dx.$$

$$\boxed{15} \quad I_{15} = \int_0^{\infty} \frac{dx}{(1+x^2)^2}.$$

$$\boxed{16} \quad I_{16} = \int_0^1 \frac{dx}{x^4-13x^2+36}.$$

$$\boxed{17} \quad I_{17} = \int \frac{\ln(\ln x)}{x} dx.$$

$$\boxed{18} \quad I_{18} = \int \frac{1+\cot x}{1-\cot x} dx.$$

$$\boxed{19} \quad I_{19} = \int \frac{\cos x + x \sin x}{x(x + \cos x)} dx.$$

$$\boxed{20} \quad I_{20} = \int_0^{\pi/2} \frac{dx}{\sin x + \sec x}.$$

$$\boxed{21} \quad I_{21} = \int_0^\infty \frac{dx}{\sqrt{1 + e^x + e^{2x}}}.$$

$$\boxed{22} \quad I_{22} = \int_0^1 x^3 e^{x^2} dx.$$

$$\boxed{23} \quad I_{23} = \int_0^1 \sqrt{1 + x \sqrt{1 + x \sqrt{1 + x \sqrt{\dots}}}} dx.$$

$$\boxed{24} \quad I_{24} = \int \left(\frac{1}{\ln x} - \frac{1}{\ln^2 x} \right) dx.$$

$$\boxed{25} \quad I_{25} = \int_1^2 (x-1)^{1/2} (2-x)^{1/2} dx.$$

2.2 2011 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int \frac{x^6 - 1}{x^4 + x^3 - x - 1} dx.$$

$$\boxed{2} \quad I_2 = \int (2 \ln x + (\ln x)^2) dx.$$

$$\boxed{3} \quad I_3 = \int \frac{2x}{\sqrt{1-x^4}} dx.$$

$$\boxed{4} \quad I_4 = \int \frac{x^2 + 1}{x + 1} dx.$$

$$\boxed{5} \quad I_5 = \int \frac{\sin^3 x + \sin^2 x - 2 \sin x - 2}{\sin^2 x + 2 \sin x + 1} dx.$$

$$\boxed{6} \quad I_6 = \int \sinh^{-2}(x) dx.$$

$$\boxed{7} \quad I_7 = \int \sec^4 x \tan^2 x dx.$$

$$\boxed{8} \quad I_8 = \int \sqrt{\csc x - \sin x} dx.$$

$$\boxed{9} \quad I_9 = \int \cos^6 x dx.$$

$$\boxed{10} \quad I_{10} = \int \frac{dx}{x^4 + 2x^2 + 1}.$$

$$\boxed{11} \quad I_{11} = \int \cos(\ln x) dx.$$

$$\boxed{12} \quad I_{12} = \int \frac{dx}{\cos x}.$$

$$\boxed{13} \quad I_{13} = \int \frac{dx}{9 \cos^2 x + 4 \sin^2 x}.$$

$$\boxed{14} \quad I_{14} = \int \frac{dx}{x^2(x^4 + 1)^{3/4}}.$$

$$\boxed{15} \quad I_{15} = \int_0^\pi \cos x \cos(3x) \cos(5x) dx.$$

$$\boxed{16} \quad I_{16} = \int \left(\frac{1}{\ln x} + \ln(\ln x) \right) dx.$$

$$\boxed{17} \quad I_{17} = \int \frac{dx}{2 + e^x}.$$

$$\boxed{18} \quad I_{18} = \int \sqrt{\frac{x}{1 - x^3}} dx.$$

$$\boxed{19} \quad I_{19} = \int \frac{4x}{1 - x^4} dx.$$

$$\boxed{20} \quad I_{20} = \int x^x (1 + \ln x) dx.$$

$$\boxed{21} \quad I_{21} = \int_0^6 \sqrt{6x - x^2} dx.$$

$$\boxed{22} \quad I_{22} = \int \sin(101x) \sin^{99} x dx.$$

$$\boxed{23} \quad I_{23} = \int x e^{e^{x^2} + x^2} dx.$$

$$\boxed{24} \quad I_{24} = \int_0^1 \frac{x^3 - 3x^2 + 3x - 1}{x^4 + 4x^3 + 6x^2 + 4x + 1} dx.$$

$$\boxed{25} \quad I_{25} = \int \sqrt{\frac{1-x}{1+x}} dx.$$

2.3 2012 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int \frac{dx}{\sqrt{x} - 1}.$$

$$\boxed{2} \quad I_2 = \int x^{1/4} \ln x dx.$$

$$\boxed{3} \quad I_3 = \int \frac{dx}{(1 + \sqrt{x}) \sqrt{x - x^2}}.$$

$$\boxed{4} \quad I_4 = \int \frac{dx}{\sqrt{x} (\sqrt[4]{x} + 1)^{10}}.$$

$$\boxed{5} \quad I_5 = \int_0^1 \sin(\cos^{-1} x) dx.$$

$$\boxed{6} \quad I_6 = \int \frac{dx}{\sqrt{1 - 4x - x^2}}.$$

$$\boxed{7} \quad I_7 = \int_{1/4}^{1/2} \left\lfloor \ln \left\lfloor \frac{1}{x} \right\rfloor \right\rfloor dx.$$

$$\boxed{8} \quad I_8 = \int_0^{\pi/2} \frac{dx}{1 + \sin x}.$$

$$\boxed{9} \quad I_9 = \int_1^{2011} \frac{\sqrt{x}}{\sqrt{2012 - x} + \sqrt{x}} dx.$$

$$\boxed{10} \quad I_{10} = \int \frac{x - 1}{(x + 1)\sqrt{x^3 + x^2 + x}} dx.$$

$$\boxed{11} \quad I_{11} = \int_{-1}^0 \frac{x^4 + 4x^3 + 6x^2 + 4x + 1}{x^3 - 3x^2 + 3x - 1} dx.$$

$$\boxed{12} \quad I_{12} = \int \left(\cos x \ln x + \frac{\sin x}{x} \right) dx.$$

$$\boxed{13} \quad I_{13} = \int \frac{dx}{x^3 - x}.$$

$$\boxed{14} \quad I_{14} = \int_0^{1/2} \frac{x \sin^{-1} x}{\sqrt{1 - x^2}} dx.$$

$$\boxed{15} \quad I_{15} = \int_0^1 x(1-x)^{99} dx.$$

$$\boxed{16} \quad I_{16} = \int_0^{\pi/2} \frac{\sin(4x)}{\sin x} dx.$$

$$\boxed{17} \quad I_{17} = \int \frac{x^{-1/2}}{1+x^{1/3}} dx.$$

$$\boxed{18} \quad I_{18} = \int \frac{dx}{\sqrt{2x^2-1}}.$$

$$\boxed{19} \quad I_{19} = \int \frac{dx}{\sqrt{e^x-1}}.$$

$$\boxed{20} \quad I_{20} = \int \frac{x}{x^4+4} dx.$$

$$\boxed{21} \quad I_{21} = \int \frac{2dx}{(\cos x - \sin x)^2}.$$

$$\boxed{22} \quad I_{22} = \int \frac{x \cosh x}{\sinh^2 x} dx.$$

$$\boxed{23} \quad I_{23} = \int_0^2 x^5 \sqrt{1+x^3} dx.$$

$$\boxed{24} \quad I_{24} = \int_0^1 \frac{x^7-1}{\ln x} dx.$$

$$\boxed{25} \quad I_{25} = \int \sqrt{\csc x - \sin x} dx.$$

2.4 2013 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int (\ln(x^2) - 2 \ln(2x)) \, dx.$$

$$\boxed{2} \quad I_2 = \int_{-1}^3 e^{|x|} dx.$$

$$\boxed{3} \quad I_3 = \int \frac{\ln x \cos x - (\sin x/x)}{\ln^2 x} dx.$$

$$\boxed{4} \quad I_4 = \int_1^{11} (x^3 - 3x^2 + 3x - 1) \, dx.$$

$$\boxed{5} \quad I_5 = \int_0^2 \sqrt{12 - 3x^2} dx.$$

$$\boxed{6} \quad I_6 = \int_0^6 (x + (x - 3)^7 + \sin(x - 3)) \, dx.$$

$$\boxed{7} \quad I_7 = \int \sin x \sqrt{1 + \tan^2 x} dx.$$

$$\boxed{8} \quad I_8 = \int \frac{x^5 - x^3 + x^2 - 1}{x^4 - x^3 + x - 1} dx.$$

$$\boxed{9} \quad I_9 = \int_0^1 \ln x dx.$$

$$\boxed{10} \quad I_{10} = \int \frac{dx}{1 - e^{-x}}.$$

$$\boxed{11} \quad I_{11} = \int_0^\pi \sin^2 x \cos^2 x dx.$$

$$\boxed{12} \quad I_{12} = \int_0^{441} \frac{\pi \sin(\pi \sqrt{x})}{\sqrt{x}} dx.$$

$$\boxed{13} \quad I_{13} = \int \tan^2 x dx.$$

$$\boxed{14} \quad I_{14} = \int_0^{256} (x - \lfloor x \rfloor)^2 dx.$$

$$\boxed{15} \quad I_{15} = \int e^{\sqrt[4]{x}} dx.$$

$$\boxed{16} \quad I_{16} = \int \cos x \cot x dx.$$

$$\boxed{17} \quad I_{17} = \int (2 \ln x + \ln^2 x) dx.$$

$$\boxed{18} \quad I_{18} = \int \frac{x^3}{1+x^2} dx.$$

$$\boxed{19} \quad I_{19} = \int \frac{dx}{2-2x+x^2}.$$

$$\boxed{20} \quad I_{20} = \int \sin x \ln(\sin x) dx.$$

$$\boxed{21} \quad I_{21} = \int \frac{x}{1-x^4} dx.$$

$$\boxed{22} \quad I_{22} = \int \sqrt{12-3x^2} dx.$$

$$\boxed{23} \quad I_{23} = \int \sec^5 x \tan^3 x dx.$$

$$\boxed{24} \quad I_{24} = \int_{-\pi/4}^{\pi/4} \frac{dx}{1 - \sin x}.$$

$$\boxed{25} \quad I_{25} = \int \frac{dx}{x\sqrt{x^2 - 2}}.$$

2.5 2014 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int_1^e \ln(x^2) dx.$$

$$\boxed{2} \quad I_2 = \int_{-9}^9 \sin(\sqrt[3]{x}) dx.$$

$$\boxed{3} \quad I_3 = \int_0^\infty \left(\frac{d}{dx} \left(e^{1+x-x^2} \right) \right) dx.$$

$$\boxed{4} \quad I_4 = \int_0^2 \sqrt{x + \sqrt{x + \sqrt{x + \sqrt{x + \dots}}}} dx.$$

$$\boxed{5} \quad I_5 = \int \sqrt{x} e^{\sqrt{x}} dx.$$

$$\boxed{6} \quad I_6 = \int \sin(2x) \cos(3x) dx.$$

$$\boxed{7} \quad I_7 = \int_0^{2\pi} |1 + 2 \sin x| dx.$$

$$\boxed{8} \quad I_8 = \int x(1-x)^{2014} dx.$$

$$\boxed{9} \quad I_9 = \int \operatorname{arsinh} x dx.$$

$$\boxed{10} \quad I_{10} = \int_{-1}^0 \frac{x^2}{x-1} dx.$$

$$\boxed{11} \quad I_{11} = \int x \arctan x dx.$$

$$\boxed{12} \quad I_{12} = \int \frac{dx}{x^2 - 15x - 2014}.$$

$$\boxed{13} \quad I_{13} = \int e^x (\ln(1+x^2) - 2(1+x) \arctan x) dx.$$

$$\boxed{14} \quad I_{14} = \int (\arcsin x)^2 dx.$$

$$\boxed{15} \quad I_{15} = \int \frac{\sqrt{x^2-1}}{x} dx.$$

$$\boxed{16} \quad I_{16} = \int x \sec^2(4x) dx.$$

$$\boxed{17} \quad I_{17} = \int \frac{2}{6-11x+6x^2-x^3} dx.$$

$$\boxed{18} \quad I_{18} = \int_0^1 \frac{dx}{[1 - \log_2(1-x)]}.$$

$$\boxed{19} \quad I_{19} = \int_0^{1/\sqrt{3}} \sqrt{x + \sqrt{x^2+1}} dx.$$

$$\boxed{20} \quad I_{20} = \int_0^{5\pi/2} \frac{dx}{2 + \cos x}.$$

2.6 2015 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int (\cos^4 x - \sin^4 x) dx.$$

$$\boxed{2} \quad I_2 = \int \frac{x}{\sqrt{2+4x}} dx.$$

$$\boxed{3} \quad I_3 = \int_0^8 \frac{\cos(\sqrt{x})}{\sqrt{x}} dx.$$

$$\boxed{4} \quad I_4 = \int \sec x dx.$$

$$\boxed{5} \quad I_5 = \int_0^{\pi/2} \frac{e^{\sin x}}{\tan x \csc x} dx.$$

$$\boxed{6} \quad I_6 = \int_1^e x \ln^2 x dx.$$

$$\boxed{7} \quad I_7 = \int \frac{dx}{5+4\sqrt{x}+x}.$$

$$\boxed{8} \quad I_8 = \int (2015)^x dx.$$

$$\boxed{9} \quad I_9 = \int_0^2 \frac{x}{(x-3)(x+5)^2} dx.$$

$$\boxed{10} \quad I_{10} = \int \frac{\ln(1+\ln x)}{x} dx.$$

$$\boxed{11} \quad I_{11} = \int \sqrt{\csc x - \sin x} dx.$$

$$\boxed{12} \quad I_{12} = \int \frac{dx}{\sqrt{x^2 + 25}}.$$

$$\boxed{13} \quad I_{13} = \int_2^e \frac{\ln^2 x - 1}{x \ln^2 x} dx.$$

$$\boxed{14} \quad I_{14} = \int e^{3x} \arctan(e^x) dx.$$

$$\boxed{15} \quad I_{15} = \int_0^4 \frac{|x-1|}{|x-2| + |x-3|} dx.$$

$$\boxed{16} \quad I_{16} = \int_0^{2\pi} \frac{dx}{\sin^4 x + \cos^4 x}.$$

$$\boxed{17} \quad I_{17} = \int \frac{1 + e^x}{1 - e^x} dx.$$

$$\boxed{18} \quad I_{18} = \int \tan^4 x dx.$$

$$\boxed{19} \quad I_{19} = \int \sin x \tan^2 x dx.$$

$$\boxed{20} \quad I_{20} = \int \frac{x+1}{x^2 + 2x + 3} dx.$$

2.7 2016 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int \tanh x dx.$$

$$\boxed{2} \quad I_2 = \int_{-4}^4 |x^3 - x| dx.$$

$$\boxed{3} \quad I_3 = \int_1^e \ln(\sqrt{x}) \, dx.$$

$$\boxed{4} \quad I_4 = \int \left(e^{e^x + e^{-x} + x} - e^{e^x + e^{-x} - x} \right) dx.$$

$$\boxed{5} \quad I_5 = \int \frac{\ln(\ln x)}{x \ln x} dx.$$

$$\boxed{6} \quad I_6 = \int_0^{\pi/3} \frac{dx}{1 + \tan^2 x}.$$

$$\boxed{7} \quad I_7 = \int_{-27}^{27} \arcsin(\sqrt[3]{x}/3) \, dx.$$

$$\boxed{8} \quad I_8 = \int_{50}^{100} \lfloor \log_2 x \rfloor \, dx.$$

$$\boxed{9} \quad I_9 = \int (e^x \cos x - e^x \sin x) \, dx.$$

$$\boxed{10} \quad I_{10} = \int_0^\infty x^3 e^{-x^2} \, dx.$$

$$\boxed{11} \quad I_{11} = \int \left((2xe^{x^2} + 1) \cos x - (e^{x^2} + x) \sin x \right) dx.$$

$$\boxed{12} \quad I_{12} = \int \left(1 + x^{1/2} + x^{1/3} \right) \left(1 + x^{-1/2} + x^{-1/3} \right) dx.$$

$$\boxed{13} \quad I_{13} = \int \sin(\sin x) \cos(\sin x) \cos x \, dx.$$

$$\boxed{14} \quad I_{14} = \int \left(\frac{\cos x + \sin x}{x^2} + \frac{\sin x - \cos x}{x} \right) dx.$$

$$\boxed{15} \quad I_{15} = \int x^3 \sqrt{x^2 + 1} dx.$$

$$\boxed{16} \quad I_{16} = \int \frac{x}{x^4 + x^2 + 1} dx.$$

$$\boxed{17} \quad I_{17} = \int e^{e^{2016x} + 6048x} dx.$$

$$\boxed{18} \quad I_{18} = \int_{\pi/3}^{\pi/2} \frac{1 - \cos x}{\sin x} dx.$$

$$\boxed{19} \quad I_{19} = \int \frac{dx}{1 - x + x^2 - x^3}.$$

$$\boxed{20} \quad I_{20} = \int_0^\infty \frac{dx}{2 + \cosh x}.$$

2.8 2017 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int \frac{x^2}{\sqrt{x^3 + 2}} dx.$$

$$\boxed{2} \quad I_2 = \int_1^\infty \frac{\ln x}{x^2} dx.$$

$$\boxed{3} \quad I_3 = \int \operatorname{sech} x dx.$$

$$\boxed{4} \quad I_4 = \int x^3 e^{x^2} dx.$$

$$\boxed{5} \quad I_5 = \int_1^2 \frac{dx}{x\sqrt{x^2 - 1}}.$$

$$\boxed{6} \quad I_6 = \int_1^{\infty} \frac{dx}{x(x^2 + 1)}.$$

$$\boxed{7} \quad I_7 = \int \cosh^{-1} x dx.$$

$$\boxed{8} \quad I_8 = \int_{-\infty}^{\infty} e^{-2x^2 - 5x - 3} dx.$$

$$\boxed{9} \quad I_9 = \int \sin(\sqrt{x}) dx.$$

$$\boxed{10} \quad I_{10} = \int_0^{\infty} \frac{dx}{\left(x + \frac{1}{x}\right)^2}.$$

$$\boxed{11} \quad I_{11} = \int \frac{(2+x)e^{-x}}{x^3} dx.$$

$$\boxed{12} \quad I_{12} = \int_0^1 \frac{dx}{\sqrt{x(1-x)}}.$$

$$\boxed{13} \quad I_{13} = \int_0^{\infty} \frac{\tanh x}{e^x} dx.$$

$$\boxed{14} \quad I_{14} = \int_0^{\pi/2} \sqrt{1 + \sin x} dx.$$

$$\boxed{15} \quad \lim_{n \rightarrow \infty} I_n = ? \quad \text{where} \quad I_1 = \int_0^1 \frac{dx}{1 + \sqrt{x}}, \quad I_2 = \int_0^1 \frac{dx}{1 + \frac{1}{1 + \sqrt{x}}}, \quad \dots$$

$$\boxed{16} \quad I_{16} = \int_{-\infty}^{\infty} \frac{\sin^2(x + \pi/4)}{e^{x^2}} dx.$$

$$\boxed{17} \quad I_{17} = \int_{-\infty}^{\infty} 3x^2 (x^3 + 1)^2 e^{-x^6 - 2x^3} dx.$$

$$\boxed{18} \quad I_{18} = \int_0^{\pi/2} \frac{dx}{1 + \tan^{2017} x}.$$

$$\boxed{19} \quad I_{19} = \int e^{2x} \cos(3x) dx.$$

$$\boxed{20} \quad I_{20} = \int ((\cos x)^{\cos x + 1} \tan x) (1 + \ln(\cos x)) dx.$$

2.9 2018 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int \frac{e^x}{e^x + 2} dx.$$

$$\boxed{2} \quad I_2 = \int \sqrt{x \sqrt[3]{x \sqrt[4]{x \sqrt[5]{x} \dots}}} dx.$$

$$\boxed{3} \quad I_3 = \int_0^{2018\pi} |\sin(2018x)| dx.$$

$$\boxed{4} \quad I_4 = \int \frac{dx}{\tan x + \cot x}.$$

$$\boxed{5} \quad I_5 = \int \frac{x^5}{2 + x^{12}} dx.$$

$$\boxed{6} \quad I_6 = \int (\cos x \cosh x + \sin x \sinh x) dx.$$

$$\boxed{7} \quad I_7 = \int \frac{e^x + \cos x}{e^x + \sin x} dx.$$

$$\boxed{8} \quad I_8 = \int \sin(\cos(\sin x)) \sin(\sin x) \cos x dx.$$

$$\boxed{9} \quad I_9 = \int \frac{dx}{1 + \sin x}.$$

$$\boxed{10} \quad I_{10} = \int \frac{\cos x}{1 - \cos(2x)} dx.$$

$$\boxed{11} \quad I_{11} = \int e^x \left(\frac{1}{x} + \ln x \right) dx.$$

$$\boxed{12} \quad I_{12} = \int \tanh^2 x dx.$$

$$\boxed{13} \quad I_{13} = \int \frac{2017x^{2016} + 2018x^{2017}}{1 + x^{4034} + 2x^{4035} + x^{4036}} dx.$$

$$\boxed{14} \quad I_{14} = \int \frac{\sin(2x) - \sin^2 x}{\cos(2x) - \cos^2 x} dx.$$

$$\boxed{15} \quad I_{15} = \int \frac{dx}{x^{25/25} x^{16/25} + x^{9/25}}.$$

$$\boxed{16} \quad I_{16} = \int_0^{\pi/2} \frac{\cos x}{2 - \cos^2 x} dx.$$

$$\boxed{17} \quad I_{17} = \int \frac{dx}{(1 + x^2)^{3/2}}.$$

$$\boxed{18} \quad I_{18} = \int \frac{dx}{\sqrt{x}\sqrt{x} - x^2}.$$

$$\boxed{19} \quad I_{19} = \int \frac{x - 1}{x + x^2 \ln x} dx.$$

$$\boxed{20} \quad I_{20} = \int \csc x \sec x dx.$$

2.10 2019 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int_0^{2\pi} \tan(\cos x) dx.$$

$$\boxed{2} \quad I_2 = \int \frac{x+1}{x(x+\ln x)} dx.$$

$$\boxed{3} \quad I_3 = \int \left(e^{x+e^x} + e^{x-e^x} \right) dx.$$

$$\boxed{4} \quad I_4 = \int_{-1/2}^{1/2} \frac{dx}{1-x^2}.$$

$$\boxed{5} \quad I_5 = \int_0^2 2^{\ln x} dx.$$

$$\boxed{6} \quad I_6 = \int_{-2\pi}^{2\pi} (\cos(3x) + \sin(2x)) (-\sin(2019x) + \cos(3x)) dx.$$

$$\boxed{7} \quad I_7 = \int \cos x (\cos(\sin x)) \cos(\sin(\sin x)) dx.$$

$$\boxed{8} \quad I_8 = \int_0^\infty \frac{e^{-2019/(4t^2)}}{t^2} dt.$$

$$\boxed{9} \quad I_9 = \int \sin(\sqrt{x}) dx.$$

$$\boxed{10} \quad I_{10} = \int_0^1 \frac{\sqrt{x}}{1+x} dx.$$

$$\boxed{11} \quad I_{11} = \int_0^{2\pi} \cos x \cos(2x) \cos(3x) dx.$$

$$\boxed{12} \quad I_{12} = \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} e^{-x^{2n}} dx.$$

$$\boxed{13} \quad I_{13} = \int_0^e x^{1/\ln x} dx.$$

$$\boxed{14} \quad I_{14} = \int_0^{\pi/100} \frac{\sin(20x) + \sin(19x)}{\cos(20x) + \cos(19x)} dx.$$

$$\boxed{15} \quad I_{15} = \int (e^x \cos^2 x + e^x \sin x \cos x - e^x \sin^2 x) dx.$$

$$\boxed{16} \quad I_{16} = \int_0^{\pi/2} \frac{\sin x}{\sin(x + \frac{\pi}{4})} dx.$$

$$\boxed{17} \quad I_{17} = \int \frac{dx}{x + \sqrt[3]{x}}.$$

$$\boxed{18} \quad I_{18} = \int_0^2 x^{x^2+1} (2 \ln x + 1) dx.$$

$$\boxed{19} \quad I_{19} = \int \frac{2x^3 - 1}{x(x^3 + 1)} dx.$$

$$\boxed{20} \quad I_{20} = \int \cos(\arctan x) dx.$$

2.11 2020 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int \frac{\ln(2x)}{x \ln x} dx.$$

$$\boxed{2} \quad I_2 = \int_0^{\infty} \frac{dx}{e^x + 1}.$$

$$\boxed{3} \quad I_3 = \int_e^{e^e} \frac{\ln x \ln(\ln x)}{x} dx.$$

$$\boxed{4} \quad I_4 = \int_0^1 \ln \left(\frac{1+x}{1-x} \right) dx.$$

$$\boxed{5} \quad I_5 = \int \frac{dx}{x^2 + (x-1)^2}.$$

$$\boxed{6} \quad I_6 = \int \sqrt{x \sqrt{x \sqrt{x \sqrt{x \dots}}}} dx.$$

$$\boxed{7} \quad I_7 = \int \sin^4 x \cos^4 x (\cos x + \sin x)(\cos x - \sin x) dx.$$

$$\boxed{8} \quad I_8 = \int \ln(1+x^2) dx.$$

$$\boxed{9} \quad I_9 = \int_0^{2\pi} (\cos x)^{2020} dx.$$

$$\boxed{10} \quad I_{10} = \int \frac{2x+1}{2x^2+2x+1} dx.$$

$$\boxed{11} \quad I_{11} = \int_{1/\sqrt{2}}^1 \frac{\arcsin x}{x^3} dx.$$

$$\boxed{12} \quad I_{12} = \int_0^{\pi/2} \sin(2x) \cos(\cos x) dx.$$

$$\boxed{13} \quad I_{13} = \int_0^{2\pi} \sin(\sin x - x) dx.$$

$$\boxed{14} \quad I_{14} = \int \left(\frac{1}{x-1} + \frac{\sum_{k=0}^{2018} (k+1)x^k}{\sum_{k=0}^{2019} x^k} \right) dx.$$

$$\boxed{15} \quad I_{15} = \int_0^{\pi/2} \frac{dx}{\tan^{\sqrt{2020}} x + 1}.$$

$$\boxed{16} \quad I_{16} = \int x(1-x)^{2020} dx.$$

$$\boxed{17} \quad I_{17} = \int \frac{\sec^4 x \tan x}{\sec^4 x + 4} dx.$$

$$\boxed{18} \quad I_{18} = \int x^{2x} (2 + 2 \ln x) dx.$$

$$\boxed{19} \quad I_{19} = \int_0^1 \sqrt{1-x^2} dx.$$

$$\boxed{20} \quad I_{20} = \int_0^\infty x^5 e^{-x^4} dx.$$

2.12 2022 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int \frac{1 + \cos x}{x + \sin x} dx.$$

$$\boxed{2} \quad I_2 = \int_1^{\sqrt{3}} \frac{\arctan x + \operatorname{arccot} x}{x} dx.$$

$$\boxed{3} \quad I_3 = \int_0^{2022} (x^2 - [x][x]) dx.$$

$$\boxed{4} \quad I_4 = \int \frac{\sinh x}{\cosh x - \sinh x} dx.$$

$$\boxed{5} \quad I_5 = \int \frac{x}{\sqrt{x-1} + \sqrt{x+1}} dx.$$

$$\boxed{6} \quad I_6 = \int_0^\pi \cos(x + \cos x) dx.$$

$$\boxed{7} \quad I_7 = \int x^3 \sin(x^2) dx.$$

$$\boxed{8} \quad I_8 = \int \frac{x}{1-x^4} dx.$$

$$\boxed{9} \quad I_9 = \int \frac{dx}{\cosh^2 x}.$$

$$\boxed{10} \quad I_{10} = \int_0^1 \left(e^{e^x} - e^{e^x - x} \right) dx.$$

$$\boxed{11} \quad I_{11} = \lim_{n \rightarrow \infty} \int_0^3 \underbrace{\sin\left(\frac{\pi}{3} \sin\left(\frac{\pi}{3} \sin\left(\dots \sin\left(\frac{\pi}{3}x\right)\right)\right)\right)}_{n \text{ sin's}} dx.$$

$$\boxed{12} \quad I_{12} = \int_0^1 \sqrt{1 - \sqrt{x}} dx.$$

$$\boxed{13} \quad I_{13} = \int \frac{x^3}{1+x+\frac{x^2}{2}+\frac{x^3}{6}} dx.$$

$$\boxed{14} \quad I_{14} = \int (\sin(x + \sin x) - \sin(x - \sin x)) dx.$$

$$\boxed{15} \quad I_{15} = \int (\tan^4 x \sec^3 x + \tan^2 x \sec^5 x) dx.$$

$$\boxed{16} \quad I_{16} = \int (1 + \ln x) \ln(\ln x) dx.$$

$$\boxed{17} \quad I_{17} = \int \left(\frac{1}{1 + \sin x} + \frac{1}{1 + \cos x} + \frac{1}{1 + \tan x} + \frac{1}{1 + \cot x} + \frac{1}{1 + \sec x} + \frac{1}{1 + \csc x} \right) dx.$$

$$\boxed{18} \quad I_{18} = \int \frac{dx}{\sqrt{x - x^2}}.$$

$$\boxed{19} \quad I_{19} = \int_0^{1/2} \left(\sum_{n=0}^{\infty} \binom{n+3}{n} x^n \right) dx.$$

$$\boxed{20} \quad I_{20} = \int \frac{dx}{1 + \cos^2 x}.$$

2.13 2023 MIT Integration Bee, Qualifying Test.

$$\boxed{1} \quad I_1 = \int x^{1/\ln x} dx.$$

$$\boxed{2} \quad I_2 = \int \operatorname{sech} x dx.$$

$$\boxed{3} \quad I_3 = \int \frac{e^x}{(1 + e^x) \ln(1 + e^x)} dx.$$

$$\boxed{4} \quad I_4 = \int (1 + x + x^2 + x^3 + x^4) (1 - x + x^2 - x^3 + x^4) dx.$$

$$\boxed{5} \quad I_5 = \int_0^4 \binom{x}{5} dx.$$

$$\boxed{6} \quad I_6 = \int (x + \sin x + x \cos x + \sin x \cos x) dx.$$

$$\boxed{7} \quad I_7 = \int (\sin^2 x + \cos^2 x + \tan^2 x + \cot^2 x + \sec^2 x + \csc^2 x) dx.$$

$$\boxed{8} \quad I_8 = \int_0^{2\pi} \lfloor 2023 \sin x \rfloor dx.$$

$$\boxed{9} \quad I_9 = \int (1 + 2 \ln x) e^{(\ln x)^2} dx.$$

$$\boxed{10} \quad I_{10} = \int \left((1-x)^3 + (x-x^2)^3 + (x^2-1)^3 - 3(1-x)(x-x^2)(x^2-1) \right) dx.$$

$$\boxed{11} \quad I_{11} = \int_{-2023}^{2023} \underbrace{||| |x| - 1| - 1| \dots | - 1|}_{2023(-1)^s} dx.$$

$$\boxed{12} \quad I_{12} = \int (\sin^6 x + \cos^6 x + 3 \sin^2 x \cos^2 x) dx.$$

$$\boxed{13} \quad I_{13} = \int (x + e + 1) x^e e^x dx.$$

$$\boxed{14} \quad I_{14} = \int_0^1 \left(\frac{x^2}{2-x^2} + \sqrt{\frac{2x}{x+1}} \right) dx.$$

$$\boxed{15} \quad I_{15} = \int \frac{1 + 2x^{2022}}{x + x^{2023}} dx.$$

$$\boxed{16} \quad I_{16} = \int (3 \sin(20x) \cos(23x) + 20 \sin(43x)) dx.$$

$$\boxed{17} \quad I_{17} = \int_0^1 \prod_{k=0}^{\infty} \left(\frac{1}{1+x^{2^k}} \right) dx.$$

$$\boxed{18} \quad I_{18} = \int \frac{\sin x}{2e^x + \cos x + \sin x} dx.$$

$$\boxed{19} \quad I_{19} = \int \frac{\ln(x/\pi)}{(\ln x)^{\ln(e\pi)}} dx.$$

$$\boxed{20} \quad I_{20} = \int_{-3/2}^{-1/2} (x^5 + 5x^4 + 10x^3 + 8x^2 + x) dx.$$

Chapter 3

The Solutions to the 2010 MIT Integration Bee, Qualifying Test

$$I_1 = \int_0^{\pi/2} \sin x \sin(2x) \sin(3x) dx$$

Solution.

For the given integral we can write the following

$$\begin{aligned} I_1 &= -\frac{1}{2} \int_0^{\pi/2} (\cos(3x) - \cos x) \sin(3x) dx \\ &= -\frac{1}{2} \int_0^{\pi/2} (\sin(3x) \cos(3x) - \sin(3x) \cos x) dx \\ &= -\frac{1}{2} \int_0^{\pi/2} \left(\frac{1}{2} \sin(6x) - \frac{1}{2} (\sin(4x) + \sin(2x)) \right) dx \\ &= -\frac{1}{4} \int_0^{\pi/2} (\sin(6x) - \sin(4x) - \sin(2x)) dx \\ &= -\frac{1}{4} \left[-\frac{1}{6} \cos(6x) + \frac{1}{4} \cos(4x) + \frac{1}{2} \cos(2x) \right]_0^{\pi/2} \\ &= -\frac{1}{4} \left(-\frac{2}{3} \right) = \frac{1}{6}. \end{aligned}$$

$$I_2 = \int_0^{\pi/2} \sin^3(2x) \cos x dx$$

Solution.

For the given integral we can write the following

$$\begin{aligned} I_2 &= \int_0^{\pi/2} (2 \sin x \cos x)^3 \cos x dx = 8 \int_0^{\pi/2} \sin^3 x \cos^4 x dx \\ &= 8 \int_0^{\pi/2} \cos^4 x \sin^2 x \sin x dx = 8 \int_0^{\pi/2} \cos^4 x (1 - \cos^2 x) \sin x dx. \end{aligned}$$

For the last integral, let us use the substitution $\cos x = t$, so that $\sin x dx = -dt$. If $x = 0$, then $t = 1$, and if $x = \pi/2$, then $t = 0$. With these new values, we get the following

$$\begin{aligned} I_2 &= -8 \int_1^0 t^4 (1 - t^2) dt = 8 \int_0^1 (t^4 - t^6) dt = 8 \left[\frac{t^5}{5} - \frac{t^7}{7} \right]_0^1 \\ &= 8 \left(\frac{1}{5} - \frac{1}{7} \right) = \frac{16}{35}. \end{aligned}$$

$$I_3 = \int (x+1)^2 (x-1)^{1/3} dx$$

Solution 1.

For the given integral we can write the following

$$\begin{aligned} I_3 &= \int \left(x^2 (x-1)^{1/3} + 2x(x-1)^{1/3} + (x-1)^{1/3} \right) dx \\ &= \int (x^2 - 1 + 1)(x-1)^{1/3} dx + 2 \int (x-1+1)(x-1)^{1/3} dx \\ &\quad + \int (x-1)^{1/3} dx \\ &= \int (x-1)(x+1)(x-1)^{\frac{1}{3}} dx + \int (x-1)^{\frac{1}{3}} dx + 2 \int (x-1)(x-1)^{\frac{1}{3}} dx \\ &\quad + 2 \int (x-1)^{\frac{1}{3}} dx + \int (x-1)^{\frac{1}{3}} dx \\ &= \int (x-1+2)(x-1)^{4/3} dx + 2 \int (x-1)^{4/3} dx + 4 \int (x-1)^{1/3} dx \\ &= \int (x-1)^{7/3} dx + 4 \int (x-1)^{4/3} dx + 4 \int (x-1)^{1/3} dx \\ &= \frac{3}{10} (x-1)^{10/3} + \frac{12}{7} (x-1)^{7/3} + 3(x-1)^{4/3} + C \\ &= 3(x-1)^{4/3} \left(\frac{1}{10} (x-1)^2 + \frac{4}{7} (x-1) + 1 \right) + C, \end{aligned}$$

where $C \in \mathbb{R}$.

Solution 2.

By using the integration by parts (tabular integration), we find

	$f(x)$		$g'(x)$	
derivatives of $f(x)$	$(x+1)^2$		$(x-1)^{1/3}$	integrations of $g'(x)$
		(+)		
	$2(x+1)$		$\frac{3}{4}(x-1)^{4/3}$	
		(-)		
	2		$\frac{3}{4} \frac{3}{7}(x-1)^{7/3}$	
		(+)		
	0		$\frac{3}{4} \frac{3}{7} \frac{3}{10}(x-1)^{10/3}$	

$$I_3 = \frac{3}{4}(x+1)^2(x-1)^{4/3} - \frac{9}{14}(x+1)(x-1)^{7/3} + \frac{27}{140}(x-1)^{10/3} + C,$$

where $C \in \mathbb{R}$.

$$I_4 = \int x \ln \left(1 + \frac{1}{x} \right) dx$$

Solution.

At the first, let us assume that $x > 0$ (Be attentive when the integral is definite!).

By using the integration by parts, let us assume

$$u = \ln \left(1 + \frac{1}{x} \right) \Rightarrow du = -\frac{dx}{x(x+1)}, \quad dv = xdx \Rightarrow v = \frac{x^2}{2}.$$

Thus, we have

$$\begin{aligned} I_6 &= \frac{x^2}{2} \ln \left(1 + \frac{1}{x} \right) + \frac{1}{2} \int \frac{x^2}{x(x+1)} dx \\ &= \frac{x^2}{2} \ln \left(1 + \frac{1}{x} \right) + \frac{1}{2} \int \left(1 - \frac{1}{1+x} \right) dx \\ &= \frac{x^2}{2} \ln \left(1 + \frac{1}{x} \right) + \frac{1}{2}x - \frac{1}{2} \ln(1+x) + C, \end{aligned}$$

where $C \in \mathbb{R}$.

$$I_5 = \int_0^1 \sin^2(\ln x) dx$$

Solution.

At the first, we note that the given integral is improper, where the integrand $f(x) = \sin^2(\ln x)$ is not continuous at $x = 0$. So, we will write

$$I_5 = \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^1 \sin^2(\ln x) dx.$$

Let us calculate the indefinite integral

$$I = \int \sin^2(\ln x) dx. \quad (3.1)$$

For this integral, we can write

$$I = \int \frac{1 - \cos(2 \ln x)}{2} dx = \frac{1}{2} \int dx - \underbrace{\frac{1}{2} \int \cos(2 \ln x) dx}_{:=J}.$$

For the integral J , by using the integration by parts, let us assume

$$u = \cos(2 \ln x) \implies du = -\frac{2 \sin(2 \ln x)}{x} dx, \quad dv = dx \implies v = x.$$

Thus, we have

$$J = x \cos(2 \ln x) + 2 \int \sin(2 \ln x) dx.$$

Now, for the last integral, by using the integration by parts, let us assume

$$u = \sin(2 \ln x) \implies du = \frac{2 \cos(2 \ln x)}{x} dx, \quad dv = dx \implies v = x.$$

Thus, we have

$$J = x \cos(2 \ln x) + 2 \left(x \sin(2 \ln x) - 2 \underbrace{\int \cos(2 \ln x) dx}_J \right).$$

From this, we get

$$5J = x \cos(2 \ln x) + 2x \sin(2 \ln x).$$

Therefore

$$J = \frac{1}{5} x \cos(2 \ln x) + \frac{2}{5} x \sin(2 \ln x) + C,$$

where $C \in \mathbb{R}$. As a result, for the integral (3.1), we have

$$I = \frac{1}{2}x - \frac{1}{10}x \cos(2 \ln x) - \frac{1}{5}x \sin(2 \ln x) + C. \quad (3.2)$$

Therefore, for the given integral I_5 , we get the following

$$\begin{aligned} I_5 &= \lim_{\varepsilon \rightarrow 0^+} \left[\frac{1}{2}x - \frac{1}{10}x \cos(2 \ln x) - \frac{1}{5}x \sin(2 \ln x) \right]_{\varepsilon^+}^1 \\ &= \frac{1}{2} - \frac{1}{10} - \lim_{\varepsilon \rightarrow 0} \left(\frac{\varepsilon}{2} - \frac{\varepsilon}{10} \cos(2 \ln \varepsilon) - \frac{\varepsilon}{5} \sin(2 \ln \varepsilon) \right) \\ &= \frac{1}{2} - \frac{1}{10} = \frac{2}{5}. \end{aligned}$$

Remark. For the integral I in (3.1). Let us use the substitution $\ln x = t$, so that $x = e^t$, and thus $dx = e^t dt$. This gives the following

$$\begin{aligned} I &= \int e^t \sin^2 t dt = \frac{1}{2} \int (1 - \cos(2t)) e^t dt = \frac{1}{2} \int e^t dt - \frac{1}{2} \int e^t \cos(2t) dt \\ &= \frac{1}{2} e^t - \frac{1}{2} \left[\frac{e^t}{5} (\cos(2t) + 2 \sin(2t)) \right] + C, \end{aligned}$$

where $C \in \mathbb{R}$.

Remember that (see (1.1))

$$\int e^{ax} \cos(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \cos(bx) + b \sin(bx)) + C.$$

But $t = \ln x$, therefore we have

$$I = \frac{1}{2}x - \frac{1}{10}x \cos(2 \ln x) - \frac{1}{5}x \sin(2 \ln x) + C,$$

which is the same as in (3.2)!

$$I_6 = \int \frac{dx}{1 + 3e^x}$$

Solution.

For the given integral we can write the following

$$\begin{aligned} I_6 &= \int \frac{dx}{e^x (3 + e^{-x})} = \int \frac{e^{-x}}{3 + e^{-x}} dx = - \int \frac{-e^{-x}}{3 + e^{-x}} dx \\ &= -\ln(3 + e^{-x}) + C, \end{aligned}$$

where $C \in \mathbb{R}$.

$$I_7 = \int_{\pi/4}^{\pi/3} \frac{dx}{\sin^3 x \cos^5 x}$$

Solution.

For the given integral we can write the following

$$\begin{aligned} I_7 &= \int_{\pi/4}^{\pi/3} \frac{1/\cos^8 x}{\sin^3 x / \cos^3 x} dx = \int_{\pi/4}^{\pi/3} \frac{1/\cos^6 x}{\tan^3 x \cos^2 x} dx \\ &= \int_{\pi/4}^{\pi/3} \frac{(1/\cos^2 x)^3}{\tan^3 x \cos^2 x} dx = \int_{\pi/4}^{\pi/3} \frac{(1 + \tan^2 x)^3}{\tan^3 x \cos^2 x} dx. \end{aligned}$$

For the last integral, let us use the substitution $\tan x = t$, so that $\frac{dx}{\cos^2 x} = dt$. If $x = \pi/4$, then $t = 1$, and if $x = \pi/3$, then $t = \sqrt{3}$. With these new values, we get the following

$$\begin{aligned} I_7 &= \int_1^{\sqrt{3}} \frac{(1+t^2)^3}{t^3} dt = \int_1^{\sqrt{3}} \frac{t^6 + 4t^4 + 3t^2 + 1}{t^3} dt \\ &= \int_1^{\sqrt{3}} \left(t^3 + 3t + \frac{3}{t} + \frac{1}{t^3} \right) dt \\ &= \left[\frac{t^4}{4} + \frac{3}{2}t^2 + 3 \ln |t| - \frac{1}{2t^2} \right]_1^{\sqrt{3}} = \frac{16}{3} + 3 \ln(\sqrt{3}). \end{aligned}$$

$$I_8 = \int_1^{\infty} \frac{dx}{x\sqrt{x^4-1}}$$

Solution 1.

For the given integral we can write the following

$$I_8 = \lim_{\substack{a \rightarrow 1^+ \\ b \rightarrow \infty}} \int_a^b \frac{dx}{x\sqrt{x^4-1}}.$$

Let us calculate the indefinite integral

$$I = \int \frac{dx}{x\sqrt{x^4-1}}. \quad (3.3)$$

For this integral, let us use the substitution $\sqrt{x^4 - 1} = t$, so that $x = (1 + t^2)^{\frac{1}{4}}$. Thus, $dx = \frac{1}{2}t(1 + t^2)^{-\frac{3}{4}}dt$. This gives the following

$$\begin{aligned} I &= \frac{1}{2} \int \frac{t(1 + t^2)^{-3/4}}{t(1 + t^2)^{1/4}} dt = \frac{1}{2} \int \frac{dt}{1 + t^2} = \frac{1}{2} \arctan(t) + C \\ &= \frac{1}{2} \arctan(\sqrt{x^4 - 1}) + C, \end{aligned}$$

where $C \in \mathbb{R}$. Therefore, for the given integral I_8 , we get

$$\begin{aligned} I_8 &= \lim_{\substack{a \rightarrow 1^+ \\ b \rightarrow \infty}} \left[\frac{1}{2} \arctan(\sqrt{x^4 - 1}) \right]_a^b \\ &= \frac{1}{2} \lim_{\substack{a \rightarrow 1^+ \\ b \rightarrow \infty}} \left(\arctan \sqrt{b^4 - 1} - \arctan \sqrt{a^4 - 1} \right) = \frac{\pi}{4}. \end{aligned}$$

Solution 2.

For the integral I in (3.3), we have

$$I = \int \frac{dx}{x\sqrt{x^4 - 1}} = \int \frac{dx}{xx^2\sqrt{1 - x^{-4}}} = \int \frac{dx}{x^3\sqrt{1 - \frac{1}{x^4}}}.$$

For the last integral, let us use the substitution $1 - \frac{1}{x^4} = t$, so that $x^{-4} = 1 - t$. Thus,

$$-4x^{-5}dx = -dt \implies \frac{dx}{x^3x^2} = \frac{dt}{4} \implies \frac{dx}{x^3} = \frac{dt}{4\sqrt{1-t}}.$$

This gives the following

$$\begin{aligned} I &= \int \frac{dt}{4\sqrt{1-t}\sqrt{t}} = \frac{1}{4} \int \frac{dt}{\sqrt{t-t^2}} = \frac{1}{4} \int \frac{dt}{\sqrt{-(t^2 - t + \frac{1}{4} - \frac{1}{4})}} \\ &= \frac{1}{4} \int \frac{dt}{\sqrt{\frac{1}{4} - (t - \frac{1}{2})^2}} = \frac{1}{4} \arcsin(2t - 1) + C = \frac{1}{4} \arcsin\left(1 - \frac{2}{x^4}\right) + C, \end{aligned}$$

where $C \in \mathbb{R}$. Therefore, for the integral I_8 , we find

$$\begin{aligned} I_8 &= \lim_{\substack{a \rightarrow 1^+ \\ b \rightarrow \infty}} \left[\frac{1}{4} \arcsin\left(1 - \frac{2}{x^4}\right) \right]_a^b \\ &= \frac{1}{4} \left(\lim_{b \rightarrow \infty} \arcsin\left(1 - \frac{2}{b^4}\right) - \arcsin\left(1 - \frac{2}{a^4}\right) \right) \\ &= \frac{1}{4} (\arcsin(1) - \arcsin(-1)) = \frac{1}{4} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = \frac{\pi}{4}. \end{aligned}$$

$$I_9 = \int \frac{dx}{x(x^5 + 1)}$$

Solution.

For the given integral we can write the following

$$I_9 = \int \frac{dx}{xx^5 \left(1 + \frac{1}{x^5}\right)} = \int \frac{dx}{x^6 \left(1 + \frac{1}{x^5}\right)}.$$

For the last integral, let us use the substitution $1/x^5 = t$, so that $dx/x^6 = -dt/5$. This gives the following

$$\begin{aligned} I_9 &= -\frac{1}{5} \int \frac{dt}{1+t} = -\frac{1}{5} \ln|1+t| + C = -\frac{1}{5} \ln \left| 1 + \frac{1}{x^5} \right| + C \\ &= -\frac{1}{5} \ln \left| \frac{x^5 + 1}{x^5} \right| + C = \frac{1}{5} \ln \left| \frac{x^5}{x^5 + 1} \right| + C, \end{aligned}$$

where $C \in \mathbb{R}$.

$$I_{10} = \int_0^{\pi/4} \sqrt{\tan x} dx$$

Solution.

Let us use the substitution $\tan x = t$, so that $dx = \frac{dt}{1+t^2}$. If $x = 0$, then $t = 0$, and if $x = \pi/4$, then $t = 1$. With these new values, we get the following

$$I_{10} = \int_0^1 \frac{\sqrt{t}}{1+t^2} dt.$$

Now, let us use the substitution $t = u^2$, so that $dt = 2u du$. If $t = 0$, then $u = 0$, and if $t = 1$, then $u = 1$. This gives the following integral

$$I_{10} = \int_0^1 \frac{2u^2}{1+u^4} du. \quad (3.4)$$

For this integral, we have

$$\begin{aligned} u^4 + 1 &= u^4 + 2u^2 + 1 - 2u^2 = (u^2 + 1)^2 - (\sqrt{2}u)^2 \\ &= (u^2 + \sqrt{2}u + 1)(u^2 - \sqrt{2}u + 1). \end{aligned}$$

Thus,

$$\frac{2u^2}{u^4 + 1} = \frac{Au + B}{u^2 + \sqrt{2}u + 1} + \frac{Cu + D}{u^2 - \sqrt{2}u + 1}. \quad (3.5)$$

For the constants A, B, C, D , we have

$$2u^2 = (Au + B) \left(u^2 - \sqrt{2}u + 1 \right) + (Cu + D) \left(u^2 + \sqrt{2}u + 1 \right). \quad (3.6)$$

By the corresponding between the sides of the equality (3.6), we find the following linear system of equations

$$\begin{cases} A + C = 0, \\ -\sqrt{2}A + B + \sqrt{2}C + D = 2, \\ A - \sqrt{2}B + C + \sqrt{2}D = 0, \\ B + D = 0. \end{cases}$$

By solving these equations, we find

$$A = -\frac{1}{\sqrt{2}}, \quad B = 0, \quad C = \frac{1}{\sqrt{2}}, \quad D = 0.$$

Therefore

$$\begin{aligned} \int \frac{2u^2}{1+u^4} du &= -\frac{1}{\sqrt{2}} \int \frac{u}{u^2 + \sqrt{2}u + 1} du + \frac{1}{\sqrt{2}} \int \frac{u}{u^2 - \sqrt{2}u + 1} du \\ &= -\frac{1}{\sqrt{2}} \cdot \frac{1}{2} \int \frac{2u + \sqrt{2} - \sqrt{2}}{u^2 + \sqrt{2}u + 1} du + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \int \frac{2u - \sqrt{2} + \sqrt{2}}{u^2 - \sqrt{2}u + 1} du \\ &= -\frac{1}{2\sqrt{2}} \int \frac{2u + \sqrt{2}}{u^2 + \sqrt{2}u + 1} du + \frac{1}{2} \int \frac{du}{u^2 + \sqrt{2}u + 1} \\ &\quad + \frac{1}{2\sqrt{2}} \int \frac{2u - \sqrt{2}}{u^2 - \sqrt{2}u + 1} du + \frac{1}{2} \int \frac{du}{u^2 - \sqrt{2}u + 1} \\ &= \frac{1}{2\sqrt{2}} \left(\int \frac{2u - \sqrt{2}}{u^2 - \sqrt{2}u + 1} du - \int \frac{2u + \sqrt{2}}{u^2 + \sqrt{2}u + 1} du \right) \\ &\quad + \frac{1}{2} \left(\int \frac{du}{\left(u + \frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} + \int \frac{du}{\left(u - \frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} \right) \\ &= \frac{1}{2\sqrt{2}} \ln \left| \frac{u^2 - \sqrt{2}u + 1}{u^2 + \sqrt{2}u + 1} \right| + \frac{1}{\sqrt{2}} \arctan(\sqrt{2}u + 1) \\ &\quad + \frac{1}{\sqrt{2}} \arctan(\sqrt{2}u - 1) + C', \end{aligned}$$

where $C' \in \mathbb{R}$. Therefore, for the integral I_{10} in (3.4), we have

$$I_{10} = \left[\frac{1}{2\sqrt{2}} \ln \left| \frac{u^2 - \sqrt{2}u + 1}{u^2 + \sqrt{2}u + 1} \right| + \frac{1}{\sqrt{2}} \arctan(\sqrt{2}u + 1) + \right.$$

$$\begin{aligned}
& + \frac{1}{\sqrt{2}} \arctan(\sqrt{2}u - 1) \Big]_0^1 \\
& = \frac{1}{2\sqrt{2}} \ln \left(\frac{2 - \sqrt{2}}{2 + \sqrt{2}} \right) + \frac{\pi}{2\sqrt{2}}.
\end{aligned}$$

Remember that

$$\arctan(a) + \arctan(b) = \arctan\left(\frac{a+b}{1-ab}\right).$$

Remark. For the integral

$$J := \int \frac{2u^2}{1+u^4} du,$$

we can write the following

$$\begin{aligned}
J &= \int \frac{2u^2}{1+u^4} du = \int \frac{2}{u^2 + \frac{1}{u^2}} du = \int \frac{1 + \frac{1}{u^2} + 1 - \frac{1}{u^2}}{u^2 + \frac{1}{u^2}} du \\
&= \underbrace{\int \frac{1 + \frac{1}{u^2}}{(u - \frac{1}{u})^2 + 2} du}_{:= J_1} + \underbrace{\int \frac{1 - \frac{1}{u^2}}{(u + \frac{1}{u})^2 - 2} du}_{:= J_2}.
\end{aligned}$$

For the integral J_1 , let us use the substitution

$$u - \frac{1}{u} = y \implies \left(1 + \frac{1}{u^2}\right) du = dy,$$

and for the integral J_2 , let we use the substitution

$$u + \frac{1}{u} = z \implies \left(1 - \frac{1}{u^2}\right) du = dz.$$

From These substitutions we get

$$\begin{aligned}
J &= \int \frac{2u^2}{1+u^4} du = \int \frac{dy}{(\sqrt{2})^2 + y^2} - \int \frac{dz}{(\sqrt{2})^2 - z^2} \\
&= \frac{1}{\sqrt{2}} \arctan(y) - \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2} + z}{\sqrt{2} - z} \right| + C \\
&= \frac{1}{\sqrt{2}} \arctan\left(u - \frac{1}{u}\right) - \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2} + u + \frac{1}{u}}{\sqrt{2} - u - \frac{1}{u}} \right| + C \\
&= \frac{1}{\sqrt{2}} \arctan\left(u - \frac{1}{u}\right) - \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2}u + u^2 + 1}{\sqrt{2}u - u^2 - 1} \right| + C,
\end{aligned}$$

where $C \in \mathbb{R}$. Therefore, for the integral I_{10} in (3.4), we have

$$\begin{aligned}
 I_{10} &= \left[\frac{1}{\sqrt{2}} \arctan \left(u - \frac{1}{u} \right) - \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2}u + u^2 + 1}{\sqrt{2}u - u^2 - 1} \right| \right]_{u \rightarrow 0}^1 \\
 &= -\frac{1}{2\sqrt{2}} \ln \left| \frac{2 + \sqrt{2}}{2 - \sqrt{2}} \right| \\
 &\quad - \lim_{u \rightarrow 0} \left(\frac{1}{\sqrt{2}} \arctan \left(u - \frac{1}{u} \right) - \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2}u + u^2 + 1}{\sqrt{2}u - u^2 - 1} \right| \right) \\
 &= \frac{1}{2\sqrt{2}} \ln \left(\frac{2 - \sqrt{2}}{2 + \sqrt{2}} \right) + \frac{\pi}{2\sqrt{2}}.
 \end{aligned}$$

$$I_{11} = \int_0^1 \frac{\ln(1+x)}{1+x^2} dx$$

Solution.

Let us use the substitution $x = \tan t$, so that $dx = dt / \cos^2 t$. If $x = 0$, then $t = 0$, and if $x = 1$, then $t = \pi/4$. With these new values, we get the following

$$I_{11} = \int_0^{\pi/4} \frac{\ln(1 + \tan t)}{1 + \tan^2 t} \frac{dt}{\cos^2 t} = \int_0^{\pi/4} \ln(1 + \tan t) dt. \quad (3.7)$$

For the last integral, let us use the substitution $y = \frac{\pi}{4} - t$, so that $dy = -dt$. If $t = 0$, then $y = \pi/4$, and if $t = \pi/4$, then $y = 0$. With these new values, we get the following

$$I_{11} = - \int_{\pi/4}^0 \ln \left(1 + \tan \left(\frac{\pi}{4} - y \right) \right) dy = \int_0^{\pi/4} \ln \left(1 + \tan \left(\frac{\pi}{4} - y \right) \right) dy.$$

But

$$\tan \left(\frac{\pi}{4} - y \right) = \frac{\tan(\pi/4) - \tan y}{1 + \tan(\pi/4) \tan y} = \frac{1 - \tan y}{1 + \tan y}.$$

Thus, we have

$$\begin{aligned}
 I_{11} &= \int_0^{\pi/4} \ln \left(1 + \frac{1 - \tan y}{1 + \tan y} \right) dy = \int_0^{\pi/4} \ln \left(\frac{2}{1 + \tan y} \right) dy \\
 &= \int_0^{\pi/4} \ln \left(\frac{2}{1 + \tan t} \right) dt.
 \end{aligned} \quad (3.8)$$

Now, by adding (3.7) and (3.8), we find

$$\begin{aligned}
 2I_{11} &= \int_0^{\pi/4} \ln(1 + \tan t) dt + \int_0^{\pi/4} \ln\left(\frac{2}{1 + \tan t}\right) dt \\
 &= \int_0^{\pi/4} \ln\left((1 + \tan t)\left(\frac{2}{1 + \tan t}\right)\right) dt \\
 &= \ln 2 \int_0^{\pi/4} dt = \frac{\pi}{4} \ln 2.
 \end{aligned}$$

Therefore

$$I_{11} = \frac{\pi}{8} \ln 2.$$

$$I_{12} = \int_{64}^{729} \frac{\sqrt{x}}{\sqrt{x} - \sqrt[3]{x}} dx$$

Solution.

Let us use the substitution $x = t^6$, so that $dx = 6t^5 dt$. If $x = 64$, then $t = 2$, and if $x = 729$, then $t = 3$. With these new values, we get the following

$$\begin{aligned}
 I_{12} &= 6 \int_2^3 \frac{t^8}{t^3 - t^2} dt = 6 \int_2^3 \frac{t^6}{t - 1} dt = 6 \int_2^3 \frac{((t^3)^2 - 1) + 1}{t - 1} dt \\
 &= 6 \int_2^3 \left(\frac{(t^3 - 1)(t^3 + 1)}{t - 1} + \frac{1}{t - 1} \right) dt \\
 &= 6 \int_2^3 \left(\frac{(t - 1)(t^2 + t + 1)(t^3 + 1)}{t - 1} + \frac{1}{t - 1} \right) dt \\
 &= 6 \int_2^3 \left((t^2 + t + 1)(t^3 + 1) + \frac{1}{t - 1} \right) dt \\
 &= 6 \int_2^3 \left(t^5 + t^4 + t^3 + t^2 + t + 1 + \frac{1}{t - 1} \right) dt \\
 &= 6 \left[\frac{t^6}{6} + \frac{t^5}{5} + \frac{t^4}{4} + \frac{t^3}{3} + \frac{t^2}{2} + t + \ln|t - 1| \right]_2^3 = \frac{10747}{10} + 6 \ln 2.
 \end{aligned}$$

$$I_{13} = \int x^x (1 + \ln x) dx \quad (3.9)$$

Solution.

For the given integral we can write the following

$$I_{13} = \int e^{x \ln x} (1 + \ln x) dx.$$

Now, let us use the substitution $x \ln x = t$, so that $(1 + \ln x)dx = dt$. This gives the following

$$I_{13} = \int e^t dt = e^t + C = e^{x \ln x} + C = e^{\ln(x^x)} + C = x^x + C,$$

where $C \in \mathbb{R}$.

$$I_{14} = \int_0^1 x^{13/2} \sqrt{1 + x^{5/2}} dx$$

Solution.

Let us use the substitution $1 + x^{5/2} = t^2$, so that $x = (t^2 - 1)^{2/5}$. Thus, $dx = \frac{4}{5}t(t^2 - 1)^{-3/5}dt$. If $x = 0$, then $t = 1$, and if $x = 1$, then $t = \sqrt{2}$. With these new values, we get the following

$$\begin{aligned} I_{14} &= \int_1^{\sqrt{2}} (t^2 - 1)^{13/5} \frac{4}{5}t(t^2 - 1)^{-3/5}dt = \frac{4}{5} \int_1^{\sqrt{2}} t^2(t^2 - 1)^2 dt \\ &= \frac{4}{5} \int_1^{\sqrt{2}} (t^6 - 2t^4 + t^2) dt = \frac{4}{5} \left[\frac{t^7}{7} - \frac{2}{5}t^5 + \frac{t^3}{3} \right]_1^{\sqrt{2}} = \frac{88\sqrt{2} - 32}{525}. \end{aligned}$$

$$I_{15} = \int_0^\infty \frac{dx}{(1 + x^2)^2}$$

Solution.

For the given integral we can write the following

$$I_{15} = \lim_{b \rightarrow \infty} \int_0^b \frac{dx}{(1 + x^2)^2}.$$

Now, let us calculate the indefinite integral

$$I = \int \frac{dx}{(1 + x^2)^2}.$$

For this integral, we can write

$$I = \int \frac{1 + x^2 - x^2}{(1 + x^2)^2} dx = \int \frac{dx}{1 + x^2} - \underbrace{\int \frac{x}{(x^2 + 1)^2} dx}_{:=J}.$$

Now, for the integral J , by using the integration by parts, let us assume

$$u = x \implies du = dx, \quad \frac{x}{(x^2 + 1)^2} dx = dv \implies v = -\frac{1}{2(1 + x^2)}.$$

Thus, we have

$$I = \int \frac{dx}{1+x^2} + \frac{x}{2(1+x^2)} - \frac{1}{2} \int \frac{dx}{1+x^2} = \frac{1}{2} \arctan x + \frac{x}{2(1+x^2)} + C,$$

where $C \in \mathbb{R}$. Therefore, for the given integral I_{15} , we get

$$I_{15} = \lim_{b \rightarrow \infty} \left[\frac{1}{2} \arctan x + \frac{x}{2(1+x^2)} \right]_0^b = \lim_{b \rightarrow \infty} \left(\frac{1}{2} \arctan b + \frac{b}{2(1+b^2)} \right) = \frac{\pi}{4}.$$

$$I_{16} = \int_0^1 \frac{dx}{x^4 - 13x^2 + 36}$$

Solution.

According to the method of partial fractions, we have

$$\begin{aligned} f(x) &:= \frac{1}{x^4 - 13x^2 + 36} = \frac{1}{(x^2 - 9)(x^2 - 4)} = \frac{1}{(x-3)(x+3)(x-2)(x+2)} \\ &= \frac{A}{x-3} + \frac{B}{x+3} + \frac{C}{x-2} + \frac{D}{x+2}. \end{aligned}$$

Where

$$\begin{aligned} A &= \frac{1}{(x+3)(x-2)(x+2)} \Big|_{x=3} = \frac{1}{30}, \\ B &= \frac{1}{(x-3)(x-2)(x+2)} \Big|_{x=-3} = -\frac{1}{30}, \\ C &= \frac{1}{(x-3)(x+3)(x+2)} \Big|_{x=2} = -\frac{1}{20}, \\ D &= \frac{1}{(x-3)(x+3)(x-2)} \Big|_{x=-2} = \frac{1}{20}. \end{aligned}$$

Thus,

$$f(x) = \frac{1}{30} \left(\frac{1}{x-3} - \frac{1}{x+3} \right) + \frac{1}{20} \left(\frac{1}{x+2} - \frac{1}{x-2} \right).$$

Therefore, for the given integral I_{16} , we have

$$\begin{aligned} I_{16} &= \left[\frac{1}{30} (\ln|x-3| - \ln|x+3|) + \frac{1}{20} (\ln|x+2| - \ln|x-2|) \right]_0^1 \\ &= \left[\frac{1}{30} \ln \left| \frac{x-3}{x+3} \right| + \frac{1}{20} \ln \left| \frac{x+2}{x-2} \right| \right]_0^1 = \frac{1}{20} \ln 3 - \frac{1}{30} \ln 2. \end{aligned}$$

$$I_{17} = \int \frac{\ln(\ln x)}{x} dx$$

Solution.

Let us use the substitution $\ln x = t$, so that $dx/x = dt$. This gives the following

$$I_{17} = \int \ln t dt.$$

By using the integration by parts, let us assume

$$u = \ln t \implies du = \frac{dt}{t}, \quad dv = dt \implies v = t.$$

Thus, we have

$$I_{17} = t \ln t - \int dt = t \ln t - t + C = \ln x (\ln(\ln x)) - \ln x + C,$$

where $C \in \mathbb{R}$.

$$I_{18} = \int \frac{1 + \cot x}{1 - \cot x} dx$$

Solution.

For the given integral we can write the following

$$I_{18} = \int \frac{1 + \frac{\cos x}{\sin x}}{1 - \frac{\cos x}{\sin x}} dx = \int \frac{\sin x + \cos x}{\sin x - \cos x} dx = \ln |\sin x - \cos x| + C,$$

where $C \in \mathbb{R}$.

$$I_{19} = \int \frac{\cos x + x \sin x}{x(x + \cos x)} dx$$

Solution 1.

For the given integral we can write the following

$$\begin{aligned} I_{19} &= \int \frac{x + \cos x + x \sin x - x}{x(x + \cos x)} dx \\ &= \int \frac{x + \cos x}{x(x + \cos x)} dx + \int \frac{x \sin x - x}{x(x + \cos x)} dx \\ &= \int \frac{dx}{x} - \int \frac{x(1 - \sin x)}{x(x + \cos x)} dx = \ln |x| - \ln |x + \cos x| + C, \end{aligned}$$

where $C \in \mathbb{R}$.

Solution 2.

For the given integral we can write the following

$$I_{19} = \int \frac{\cos x + x \sin x}{x^2 \left(1 + \frac{\cos x}{x}\right)} dx.$$

Now, let us use the substitution

$$\frac{\cos x}{x} = t \implies \frac{-x \sin x - \cos x}{x^2} dx = dt \implies \frac{x \sin x + \cos x}{x^2} dx = -dt.$$

This gives the following

$$\begin{aligned} I_{19} &= - \int \frac{dt}{1+t} = -\ln|1+t| + C = -\ln\left|1 + \frac{\cos x}{x}\right| + C \\ &= -\ln\left|\frac{x + \cos x}{x}\right| + C = \ln|x| - \ln|x + \cos x| + C, \end{aligned}$$

where $C \in \mathbb{R}$.

$$I_{20} = \int_0^{\pi/2} \frac{dx}{\sin x + \sec x}$$

Solution 1.

For the given integral we can write the following

$$I_{20} = \int_0^{\pi/2} \frac{\cos x}{1 + \sin x \cos x} dx. \quad (3.10)$$

By using the King property of integration (1.13), we find

$$I_{20} = \int_0^{\pi/2} \frac{\cos\left(\frac{\pi}{2} - x\right)}{1 + \sin\left(\frac{\pi}{2} - x\right) \cos\left(\frac{\pi}{2} - x\right)} dx = \int_0^{\pi/2} \frac{\sin x}{1 + \sin x \cos x} dx. \quad (3.11)$$

By adding (3.10) and (3.11), we find

$$\begin{aligned} 2I_{20} &= \int_0^{\pi/2} \frac{\sin x + \cos x}{1 + \sin x \cos x} dx = 2 \int_0^{\pi/2} \frac{\sin x + \cos x}{2 + 2 \sin x \cos x} dx \\ &= 2 \int_0^{\pi/2} \frac{\sin x + \cos x}{3 - (\cos x - \sin x)^2} dx. \end{aligned}$$

Now, for the last integral, let us use the substitution $\sin x - \cos x = t$, so that $(\sin x + \cos x)dx = dt$. If $x = 0$, then $t = -1$, and if $x = \pi/2$, then $t = 1$. With

these new values, we get the following

$$\begin{aligned}
 2I_{20} &= 2 \int_{-1}^1 \frac{dt}{3-t^2} = \cancel{2} \frac{1}{\cancel{2}\sqrt{3}} \left[\ln \left| \frac{\sqrt{3}+t}{\sqrt{3}-t} \right| \right]_{-1}^1 \\
 &= \frac{1}{\sqrt{3}} \left(\ln(\sqrt{3}+1) - \ln(\sqrt{3}-1) - \ln(\sqrt{3}-1) + \ln(\sqrt{3}+1) \right) \\
 &= \frac{2}{\sqrt{3}} \left(\ln(\sqrt{3}+1) - \ln(\sqrt{3}-1) \right) = \frac{2}{\sqrt{3}} \ln(2+\sqrt{3}).
 \end{aligned}$$

Therefore

$$I_{20} = \frac{1}{\sqrt{3}} \ln(2+\sqrt{3}).$$

Solution 2.

For the given integral we can write the following

$$I_{20} = \int_0^{\pi/2} \frac{\cos x}{1 + \sin x \cos x} dx = \int_0^{\pi/2} \frac{2 \cos x}{2 + \sin(2x)} dx.$$

Now, let us calculate the indefinite integral

$$J := \int \frac{2 \cos x}{2 + \sin(2x)} dx.$$

For this, we have

$$\begin{aligned}
 J &= \int \frac{\cos x + \sin x + \cos x - \sin x}{2 + \sin(2x)} dx \\
 &= \int \frac{\cos x + \sin x}{2 + \sin(2x)} dx + \int \frac{\cos x - \sin x}{2 + \sin(2x)} dx \\
 &= \underbrace{\int \frac{\cos x + \sin x}{3 - (\sin x - \cos x)^2} dx}_{:=J_1} + \underbrace{\int \frac{\cos x - \sin x}{1 + (\sin x + \cos x)^2} dx}_{:=J_2}.
 \end{aligned}$$

For J_1 , let us use the substitution $\sin x - \cos x = t$, thus $(\cos x + \sin x)dx = dt$, and for J_2 , let us use the substitution $\sin x + \cos x = u$, thus $(\cos x - \sin x)dx = du$. This gives the following

$$\begin{aligned}
 J &= \int \frac{dt}{3-t^2} + \int \frac{du}{1+u^2} = \frac{1}{2\sqrt{3}} \ln \left| \frac{\sqrt{3}+t}{\sqrt{3}-t} \right| + \arctan u + C \\
 &= \frac{1}{2\sqrt{3}} \ln \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} \right| + \arctan(\sin x + \cos x) + C,
 \end{aligned}$$

where $C \in \mathbb{R}$. Therefore, for the given integral I_{20} , we get

$$\begin{aligned}
 I_{20} &= \left[\frac{1}{2\sqrt{3}} \ln \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} \right| + \arctan(\sin x + \cos x) \right]_0^{\pi/2} \\
 &= \frac{1}{2\sqrt{3}} \ln \left(\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \right) + \frac{\pi}{4} - \frac{1}{2\sqrt{3}} \ln \left(\frac{\sqrt{3} - 1}{\sqrt{3} + 1} \right) - \frac{\pi}{4} \\
 &= \frac{1}{2\sqrt{3}} \ln \left(\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \cdot \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \right) = \frac{1}{2\sqrt{3}} \ln \left(\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \right)^2 \\
 &= \frac{1}{\sqrt{3}} \ln \left(\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \right) = \frac{1}{\sqrt{3}} \ln(2 + \sqrt{3}).
 \end{aligned}$$

$$I_{21} = \int_0^\infty \frac{dx}{\sqrt{1 + e^x + e^{2x}}}$$

Solution.

For the given integral we can write the following

$$I_{21} = \int_0^\infty \frac{dx}{e^x \sqrt{1 + e^{-x} + (e^{-x})^2}} = \int_0^\infty \frac{e^{-x}}{\sqrt{1 + e^{-x} + (e^{-x})^2}} dx.$$

For the last integral, let us use the substitution $e^{-x} = t$, so that $e^{-x} dx = -dt$. If $x = 0$, then $t = 1$, and if $x \rightarrow \infty$, then $t \rightarrow 0$. With these new values, we get the following

$$\begin{aligned}
 I_{21} &= - \int_1^0 \frac{dt}{\sqrt{t^2 + t + 1}} = \int_0^1 \frac{dt}{\sqrt{t^2 + t + \frac{1}{4} + \frac{3}{4}}} = \int_0^1 \frac{dt}{\sqrt{(t + \frac{1}{2})^2 + \frac{3}{4}}} \\
 &= \left[\ln \left| t + \frac{1}{2} + \sqrt{t^2 + t + 1} \right| \right]_0^1 = \ln \left(\frac{3 + 2\sqrt{3}}{2} \right) - \ln \left(\frac{3}{2} \right) \\
 &= \ln(3 + 2\sqrt{3}) - \ln 3.
 \end{aligned}$$

$$I_{22} = \int_0^1 x^3 e^{x^2} dx$$

Solution.

For the given integral we can write the following

$$I_{22} = \int_0^1 x x^2 e^{x^2} dx.$$

Let us use the substitution $x^2 = t$, so that $2xdx = dt$. If $x = 0$, then $t = 0$, and if $x = 1$, then $t = 1$. With these new values, we get the following

$$I_{22} = \frac{1}{2} \int_0^1 te^t dt = \frac{1}{2} \left([te^t]_0^1 - \int_0^1 e^t dt \right) = \frac{1}{2} (e - (e - 1)) = \frac{1}{2}.$$

Where we used the integration by parts, by assuming $u = t$ and $e^t dt = dv$.

$$I_{23} = \int_0^1 \sqrt{1 + x\sqrt{1 + x\sqrt{1 + x\sqrt{\dots}}}} dx$$

Solution.

Let we set

$$t = \sqrt{1 + x\sqrt{1 + x\sqrt{1 + x\sqrt{\dots}}}} > 0, \quad \forall x \in [0, 1].$$

Then

$$t^2 = 1 + x\sqrt{1 + x\sqrt{1 + x\sqrt{1 + x\sqrt{\dots}}}} = 1 + xt,$$

i.e.

$$t^2 - xt - 1 = 0.$$

By solving this quadratic equation, for t , we find the following solutions:

$$t_1 = \frac{x}{2} - \frac{1}{2}\sqrt{x^2 + 4} < 0, \quad t_2 = \frac{x}{2} + \frac{1}{2}\sqrt{x^2 + 4} > 0, \quad \forall x \in [0, 1].$$

Therefore, we have the following integral (for $t > 0$)

$$I_{23} = \int_0^1 \left(\frac{x}{2} + \frac{1}{2}\sqrt{x^2 + 4} \right) dx. \quad (3.12)$$

Now, let us calculate the following indefinite integral

$$I = \int \sqrt{x^2 + 4} dx.$$

For this, we have

$$I = 2 \int \sqrt{1 + (x/2)^2} dx.$$

Now, let us use the substitution $x/2 = \sinh t$, so that $dx = 2 \cosh t dt$, and $t = \operatorname{arcsinh}(x/2)$. This gives the following

$$\begin{aligned}
I &= 2 \int \sqrt{1 + \sinh^2 t} (2 \cosh t) dt = 4 \int \cosh^2 t dt = 4 \int \frac{1 + \cosh(2t)}{2} dt \\
&= 2 \left(t + \frac{1}{2} \sinh(2t) \right) + C = 2t + \sinh(2t) + C \\
&= 2 \operatorname{arcsinh} \left(\frac{x}{2} \right) + \sinh \left(2 \operatorname{arcsinh} \left(\frac{x}{2} \right) \right) + C \\
&= 2 \operatorname{arcsinh} \left(\frac{x}{2} \right) + 2 \sinh \left(\operatorname{arcsinh} \left(\frac{x}{2} \right) \right) \cosh \left(\operatorname{arcsinh} \left(\frac{x}{2} \right) \right) + C \\
&= 2 \operatorname{arcsinh} \left(\frac{x}{2} \right) + x \sqrt{1 + \left(\frac{x}{2} \right)^2} + C = 2 \operatorname{arcsinh} \left(\frac{x}{2} \right) + \frac{x}{2} \sqrt{x^2 + 4} + C,
\end{aligned}$$

where $C \in \mathbb{R}$.

As a result of the previous, from (3.12), we have

$$\begin{aligned}
I_{23} &= \int_0^1 \left(\frac{x}{2} + \frac{1}{2} \sqrt{x^2 + 4} \right) dx = \left[\frac{x^2}{4} + \operatorname{arcsinh} \left(\frac{x}{2} \right) + \frac{x}{4} \sqrt{x^2 + 4} \right]_0^1 \\
&= \frac{1 + \sqrt{5}}{4} + \operatorname{arcsinh} \left(\frac{1}{2} \right) = \frac{1 + \sqrt{5}}{4} + \ln \left(\frac{1}{2} + \sqrt{1 + \frac{1}{4}} \right) \\
&= \frac{1 + \sqrt{5}}{4} + \ln \left(\frac{1 + \sqrt{5}}{2} \right).
\end{aligned}$$

Where we used the fact

$$\operatorname{arcsinh}(x) = \ln \left(x + \sqrt{1 + x^2} \right), \quad \forall x \in \mathbb{R}.$$

$$I_{24} = \int \left(\frac{1}{\ln x} - \frac{1}{\ln^2 x} \right) dx$$

Solution 1.

We note that

$$\left(\frac{x}{\ln x} \right)' = \frac{\ln x - 1}{\ln^2 x} = \frac{1}{\ln x} - \frac{1}{\ln^2 x}, \quad \forall x \in]0, \infty[\setminus \{1\}.$$

Therefore, for the given integral, we get

$$I_{24} = \int \left(\frac{x}{\ln x} \right)' dx = \frac{x}{\ln x} + C,$$

where $C \in \mathbb{R}$.

Solution 2.

Let us use the substitution $\ln x = t$, so that $x = e^t$ and $dx = e^t dt$. This gives the following

$$I_{24} = \int \left(\frac{1}{t} - \frac{1}{t^2} \right) e^t dt = \underbrace{\int \frac{e^t}{t} dt}_{:=J} - \int \frac{e^t}{t^2} dt.$$

For the integral J , by using the integration by parts, we assume

$$u = \frac{1}{t} \implies du = -\frac{dt}{t^2}, \quad dv = e^t dt \implies v = e^t.$$

Thus,

$$I_{24} = \underbrace{\frac{e^t}{t} + \int \frac{e^t}{t^2} dt}_{J} - \int \frac{e^t}{t^2} dt = \frac{e^t}{t} + C = \frac{e^{\ln x}}{\ln x} + C = \frac{x}{\ln x} + C,$$

where $C \in \mathbb{R}$.

$$I_{25} = \int_1^2 (x-1)^{1/2} (2-x)^{1/2} dx$$

Solution.

For the given integral we can write the following

$$\begin{aligned} I_{25} &= \int_1^2 \sqrt{(x-1)(2-x)} dx = \int_1^2 \sqrt{-x^2 + 3x - 2} dx \\ &= \int_1^2 \sqrt{-\left(x^2 - 3x + \frac{9}{4} - \frac{9}{4}\right) - 2} dx = \int_1^2 \sqrt{\frac{1}{4} - \left(x - \frac{3}{2}\right)^2} dx \\ &= \frac{1}{2} \int_1^2 \sqrt{1 - (2x-3)^2} dx. \end{aligned}$$

For the last integral, let us use the substitution $2x - 3 = t$, so that $2dx = dt$. If $x = 1$, then $t = -1$, and if $x = 2$, then $t = 1$. With these new values, we get the following

$$I_{25} = \frac{1}{4} \int_{-1}^1 \sqrt{1-t^2} dt = \frac{1}{2} \int_0^1 \sqrt{1-t^2} dt,$$

where the function $t \mapsto \sqrt{1-t^2}$ is an even.

Now, let us use the substitution $t = \sin y$, so that $dt = \cos y dy$. If $t = 0$, then $y = 0$, and if $t = 1$, then $y = \pi/2$. With these new values, we get the following

$$\begin{aligned} I_{25} &= \frac{1}{2} \int_0^{\pi/2} \sqrt{1 - \sin^2 y} \cos y dy = \frac{1}{2} \int_0^{\pi/2} \cos^2 y dy \\ &= \frac{1}{4} \int_0^{\pi/2} (1 + \cos(2y)) dy = \frac{1}{4} \left[y + \frac{1}{2} \sin(2y) \right]_0^{\pi/2} \\ &= \frac{1}{4} \left(\frac{\pi}{2} + \frac{1}{2} \sin \pi - 0 \right) = \frac{\pi}{8}. \end{aligned}$$

Chapter 4

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About the book:

This book contains the solutions with some details for the qualifying tests of the MIT Integration Bee from 2010 to 2023. MIT Integration Bee is an annual integral calculus competition pioneered in 1981 at the Massachusetts Institute of Technology (MIT). In the first chapter of this book, there is a review of the basic integration formulas, and techniques such as integration by substitution, integration by parts, trigonometric and hyperbolic integrals, integrals of irrational functions, and integration of binomial differentials. In the remaining 13 chapters, detailed solutions to the integrals of each year were presented in their own chapter. The book is devoted to students preparing for the Integration Bee, teachers looking for integration problems with different levels of challenge to use in their classroom, or math enthusiasts looking to expand their knowledge and skills in integration calculus. This book will be updated periodically every year, as solutions to problems will be added for the coming years.

About the author:

Mohammad S. Alkousa earned his Ph.D. from the Moscow Institute of Physics and Technology (MIPT) in 2020 in mathematical optimization. Since 2020 he works as a researcher at MIPT. Prior to that, he got a BSc and MSc degree in mathematics at Damascus University with a dissertation in complex analysis (elliptic functions theory). Until now, he participated in more than 20 Scopus-published papers in the research group under the supervision of Prof. Alexander V. Gasnikov at MIPT. His scientific works also comprise 6 mini-books (Complex Beauties, translation from English to Arabic) and a book on the subject of real infinite series.